COMPREHENSIVE ANALYSIS OF WATER QUALITY OF RAPTI RIVER IN GORAKHPUR REGION

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by

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DECLARATION

I hereby declare that the work presented in this report entitled **"COMPREHENSIVE ANALYSIS OF WATER QUALITY OF RAPTI RIVER IN GORAKHPUR REGION"**, was carried out by us. I have not submitted the matter embodied in this report for the award of any other degree or diploma of any other University or Institute.

I have given due credit to the original sources for all the words, ideas, diagrams, graphics, computer programs, experiments, results, that are not my original contribution. I have used quotation marks to identify verbatim sentences and given credit to the original sources.

I affirm that no portion of my work is plagiarized, and the experiments and results reported in the report are not manipulated. In the event of a complaint of plagiarism and the manipulation of the experiments and results, I shall be fully responsible and answerable.

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CERTIFICATE

This is to certify that Project Report entitled "**COMPREHENSIVE ANALYSIS OF WATER QUALITY OF RAPTI RIVER IN GORAKHPUR REGION**" submitted by **AJEET YADAV,** in partial fulfilment of the requirement for the award of **Bachelor of Technology in Civil Engineering** from Buddha Institute of Technology, Gorakhpur affiliated to Dr. A. P. J. Abdul Kalam Technical University, Lucknow, Uttar Pradesh represents the work carried out by students under my supervision. The project embodies result of original work and studies carried out by the students themselves and the contents of the project do not form the basis for the award of any other degree to the candidate or to anybody else.

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ABSTRACT

A Study was carried at Gorakhpur city and its adjoining rivers to access the water quality of Rapti river in Gorakhpur city the population Industrialization increases day by day and the Garbages are dumped into rivers which is related to Rapti river. Due to this the water gets polluted with high rate. The slush of garbage is mixed in water and get polluted. In which we identified and determine the quality of water in Gorakhpur city by choosing four location such as Kalesar, Zero point and Nausad and Rajghat bridge determine the selected parameter. This report is detailed assessment of water quality parameters of Rapti river in Gorakhpur. Physical and Chemical water quality parameters are being tested and reported. Turbidity, PH, Hardness, Alkalinity and Hardness are tested.

Keywords: Rapti River, Water Quality of Gorakhpur, Pollution, Environmental Impact Assessment, Water Parameters, Water Management, River Pollution.

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1.1 INTRODUCTION

Rivers are among the most important sources of freshwater on Earth and often originate from the highest elevations in a region, providing a continuous flow of cold water downstream (UNESCO, 2020). Globally, rivers drain nearly 75% of the Earth's land surface, making them crucial for maintaining ecological balance and supporting human settlements (Shiklomanov & Rodda, 2003). In India, a country heavily dependent on agriculture, rivers serve as the primary source of irrigation and freshwater supply (Ministry of Jal Shakti, 2019). States with perennial rivers rely on them for crop cultivation, and this consistent water availability helps mitigate the uncertainties associated with irregular rainfall.

Given their significance, it is essential for the government and stakeholders to ensure rivers do not dry up. Numerous river valley projects and dams have been constructed across India to conserve water and regulate its distribution (CWC, 2018).

Water, being a vital natural resource, sustains life and maintains ecological and economic stability (WHO, 2017). However, rapid urbanization, industrialization, and unregulated waste disposal have severely degraded river water quality worldwide (UNEP, 2016). Pollution from domestic sewage, agricultural runoff, industrial effluents, and landfill leachates poses serious risks to aquatic ecosystems and human health.

The Rapti River, a major tributary of the Ghaghara River, flows through Nepal and the Indian states of Uttar Pradesh and Bihar, supporting thousands of people through irrigation, drinking water, fisheries, and domestic needs (IITR, 2021). In recent decades, anthropogenic activities such as untreated sewage discharge, industrial waste, religious activities, and extensive agricultural runoff have deteriorated the river's water quality (CPCB, 2020).

To understand pollution levels and manage river systems effectively, regular monitoring of water quality is essential. Physico-chemical parameters such as pH, dissolved oxygen (DO), biochemical oxygen demand (BOD), chemical oxygen demand (COD), turbidity, and total dissolved solids (TDS) are widely used as indicators of water health (APHA, 2017). This study assesses these parameters to evaluate the current condition of the Rapti River and compares results with standards set by the Bureau of Indian Standards (BIS 10500:2012) and the World Health Organization (WHO, 2017).

1.2 Classification of Water

Based on its source, water can be classified as surface water (rivers, lakes, ponds) or groundwater (aquifers, wells). Both types are vulnerable to contamination from domestic, industrial, and agricultural activities (EPA, 2022).

Water quality classification includes the following categories:
a. **Potable Water-** Safe for drinking.
b. **Contaminated Water** – Contains unwanted chemicals or substances.
c. **Polluted Water** – Water whose quality is significantly degraded beyond safe limits.
d. **Infected Water** – Contains disease-causing microorganisms.

CPCB further classifies water into five categories (Class A–E) based on Designated Best Use (CPCB, 2018).

- Class A: Drinking water after disinfection
- Class B: Outdoor bathing
- Class C: Drinking water after conventional treatment
- Class D: Propagation of wildlife and fisheries
- Class E: Irrigation and industrial cooling

Water quality may also be assessed by total dissolved solids (TDS). TDS below 300 mg/L indicates excellent water, whereas levels above 1200 mg/L indicate poor quality (WHO, 2017). Similarly, chloride levels below 250 mg/L are considered acceptable for drinking, while higher values indicate contamination (BIS 10500:2012).

1.3 Background on Water Pollution and Its Significance

Water pollution occurs when harmful substances—chemical, biological, or physical—enter water bodies, causing deterioration in quality (UNEP, 2016). Major pollutants include heavy metals, pesticides, sewage, industrial effluents, and pathogenic bacteria.

Polluted water can cause diseases like cholera, hepatitis, and dysentery (WHO, 2017). It also disrupts aquatic ecosystems by reducing DO and harming flora and fauna. In India, where rivers serve as a major freshwater source, maintaining water quality is vital for health and socio-economic development (MoEFCC, 2020).

Continuous monitoring helps identify pollution hotspots, evaluate treatment effectiveness, and guide policy decisions for sustainable water management.

1.4 Importance of River Water Quality Monitoring

Rivers are dynamic ecosystems that respond quickly to environmental and anthropogenic changes. Monitoring helps evaluate pollution trends, detect contamination sources, and ensure compliance with standards such as CPCB, BIS, and WHO guidelines (CPCB, 2020). It also supports ecological protection, early hazard detection, and sustainable river basin management.

For rivers like the Rapti, which face stress from untreated sewage, agricultural runoff, and seasonal variations, monitoring is essential to identify vulnerable stretches and adopt effective restoration strategies (IITR, 2021).

1.5 Quality of Water

a. Raw Water

Raw water refers to water that has not undergone any artificial treatment (APHA, 2017).

b. Water Quality Parameters

Water quality parameters include physical, chemical, and biological characteristics that determine suitability for various purposes.

1.5.1 Physical Parameters

1.5.1.1 Turbidity

Turbidity is the measure of water cloudiness due to suspended particles such as silt, clay, organic matter, and microorganisms. It is measured in NTU using a nephelometer (APHA, 2017). High turbidity reduces light penetration, affects photosynthesis, and indicates the presence of contaminants. It is strongly influenced by suspended solids and TDS levels (EPA, 2022).

indicator of suspended sediments in water, the presence of pathogens, bacteria, and other contaminants Measurement of turbidity: Turbidity of the sample can be measured by-

a. Turbidity Rod Method

b. Jackson Turbid Meter Method

c. Baylis Turbidity & Nephlometer

Limit of Turbidity: Acceptable limit = 1NTU Cause for Rejection = 10 NTU

- **Sources of Turbidity**

- Soil erosion from construction, agriculture, or deforestation, especially during rainfall.
- Urban runoff carrying sediments, oils, and other pollutants into rivers.
- Domestic sewage and industrial discharge containing organic and inorganic particles.
- Algal blooms and microbial growth, often caused by nutrient pollution (especially nitrogen and phosphorus).
- Resuspension of sediments from the riverbed due to water flow or human activities like dredging.

- **Effects of High Turbidity**

1. Reduces sunlight penetration, affecting aquatic plants that rely on photosynthesis.
2. Raises water temperature, as more suspended particles absorb heat.
3. Lowers dissolved oxygen (DO) due to reduced photosynthesis and increased biological activity.
4. Harms aquatic organisms, especially fish, by clogging gills and reducing feeding efficiency.
5. Increases risk of pathogens, as bacteria and viruses often attach to suspended particles.
6. Affects water treatment, making filtration and disinfection more difficult and costly.

- **Desirable Turbidity Levels**

According to the **Bureau of Indian Standards (IS 10500: 2012)** for drinking water:

- **Desirable limit**: 1 NTU
- **Maximum permissible limit**: 5 NTU

However, for river water used for various purposes (like irrigation or bathing), permissible turbidity levels may vary based on environmental and local regulatory standards.

1.5.1.2 Temperature:

The effect both chemical and biological reaction in water an average increase of 10 degree Celsius in the temp. of water. It almost the double the biological activities for water supplies temp. should be in the range of 10 to 25 degree Celsius. Temperature measures the kinetic energy of molecules in the water, measured in degrees Fahrenheit or Celsius. Temperature is one of the most essential parameters of water quality, as it influences biological activity and growth in water, plus it has a direct effect on the water chemistry, influencing the other main water quality parameters.

Temperature is one of the most fundamental physical parameters in water quality analysis. It has a significant influence on chemical, biological, and physical processes within a water body.

Even small changes in water temperature can have a major impact on aquatic ecosystems, making it a critical indicator for assessing the environmental health of rivers like the Rapti River.

➢ **Sources of Temperature Variation in Rivers**

- Natural causes such as seasonal changes and sunlight intensity.
- Discharge of heated water from industries or thermal power plants (thermal pollution).
- Removal of vegetation along riverbanks, which reduces shade and increases heat absorption.
- Urban runoff from paved surfaces that heat up during the day and transfer warm water into rivers.
- Reduced flow rate, which gives water more time to absorb heat.

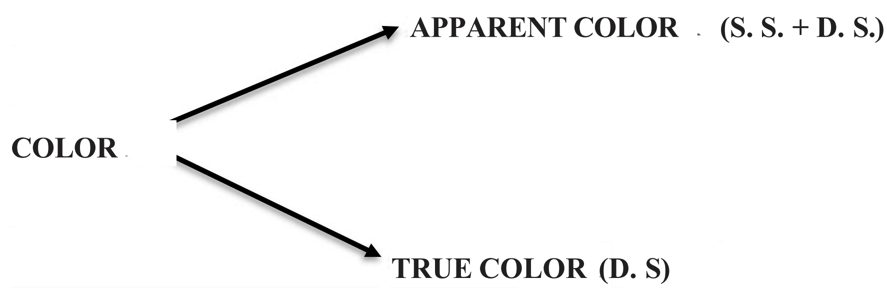
➢ **Optimal and Harmful Temperature Ranges**

- Ideal range for most freshwater aquatic life: 15°C to 30°C
- Stressful or potentially harmful range: > 35°C
- Rapid fluctuations (even within the safe range) can still stress aquatic organisms.

➢ **Measurement and Units**

Water temperature is measured using a thermometer or temperature probe, typically in degrees Celsius (°C). For accurate water quality assessment, temperature should be measured in situ (directly at the site) as it can change quickly once water is removed from its source.

1.5.1.3 Color



Water used in households should be colourless under normal conditions. If tap water appears blue or green, it often indicates contamination by microscopic organisms, algae, or suspended particles (WHO, 2017). Stagnant water bodies such as swamps commonly develop colour due to decaying organic matter, which releases humic and fulvic substances (APHA, 2017). Water colour is usually assessed by comparing a sample to standard color solutions or colour glass disks.

Some organic compounds responsible for color can react with chlorine during disinfection to form **carcinogenic compounds**, including trihalomethanes (THMs) and haloacetic acids (HAAs) (EPA, 2022).

Color is one of the most basic and easily observed physical indicators of water quality. While pure water is naturally colourless, the presence of dissolved or suspended substances—including metals, industrial chemicals, or biological materials—can impart noticeable hues

(UNEP, 2016).

- **Types of Color in Water**

1. True Color

True color results from dissolved substances such as natural organic matter or dissolved metals and is measured after filtration. It is often associated with humic acids from decaying vegetation or industrial discharges (APHA, 2017).

2. Apparent Color

Apparent color includes contributions from both dissolved materials and suspended particles such as algae, clay, or silt. This is the color observed directly in unfiltered water (EPA, 2022).

- **Common Causes of Color in River Water**

• Decaying organic matter releasing tannins and humic substances (yellow–brown color) (UNESCO, 2020).
• Industrial effluents containing dyes, chemicals, and heavy metals (CPCB, 2020).
• Algal blooms causing green, blue-green, or red coloration in nutrient-rich waters (WHO, 2017).
• Domestic sewage inputs causing greyish or yellowish coloration (MoEFCC, 2020).
• Iron and manganese contamination causing reddish or blackish colors (BIS 10500:2012).
• Erosion-derived clay and silt increasing turbidity and causing a muddy appearance (EPA, 2022).

- **Measurement of Color**

Color is measured in **Platinum-Cobalt Units (PCU)** or **Hazen Units** using a color comparator or spectrophotometer (APHA, 2017).

- **BIS IS 10500:2012 drinking water limits:**

- **Desirable limit:** 5 PCU

- **Permissible limit:** 15 PCU (in absence of alternate source)

Though color does not always indicate toxicity, it typically signals the presence of pollutants affecting water quality and aesthetics (WHO, 2017).

➢ **Effects of Colored Water**

• Reduces aesthetic and recreational value (BIS 10500:2012).
• Indicates potential pollution from sewage/industrial discharge (CPCB, 2020).
• May react with chlorine to form harmful disinfection by-products (EPA, 2022).
• Reduces light penetration, affecting photosynthesis and aquatic ecosystems (UNEP, 2016).

1.5.1.4 Total Dissolved Solids (TDS)

TDS enters water from sewage, agricultural runoff, industrial effluents, and treatment chemicals (EPA, 2022). The higher the TDS, the greater the electrical conductivity of the water.

TDS represents the total concentration of dissolved inorganic salts (Ca^{2+}, Mg^{2+}, Na^{+}, K^{+}, Cl^{-}, SO_4^{2-}) and dissolved organic matter (APHA, 2017). It is expressed in **mg/L or ppm** and affects water taste, corrosiveness, and suitability for drinking, irrigation, and industry.

WHO guidelines classify water with:

<300 mg/L: Excellent

>1200 mg/L: Poor/unacceptable (WHO, 2017)

1.5.1.5 Taste and Odor

Taste and odor problems may arise from inorganic salts such as sodium, magnesium, calcium, and potassium, or from organic matter, including algal metabolites (WHO, 2017). Algae often release oils and volatile compounds (e.g., geosmin, MIB) that cause earthy or musty odors (EPA, 2022).

Some taste- and odor-causing organic compounds can react with chlorine to form carcinogenic disinfection by-products, such as THMs (EPA, 2022).

Taste and odor are measured using an **Osmoscope**, expressed as TON (Threshold Odor Number**)** (APHA, 2017).

- **Acceptability Standard:**

Acceptable: 1 TON

Case for Rejection point: 3 TON (BIS 10500:2012)

- **Impact of Taste and Odor on Water Quality**

• **Aesthetic Decline:** Unpleasant taste/odor reduces water acceptability (WHO, 2017).
• **Consumer Confidence:** People may doubt water safety even when chemically safe.
• **Indicator of Pollution:** Strong odors may signal contamination from sewage, industrial chemicals, or algal blooms (CPCB, 2020).

- **Measurement of Taste and Odor**

Taste and odor are mainly evaluated through sensory analysis by trained personnel (APHA, 2017).

Taste is assessed for bitterness, saltiness, metallic, or chemical flavors.

Odor is assessed for earthy, musty, sulfurous, fishy, or chemical smells.

Water treatment plants commonly use **activated carbon**, **aeration**, or **ozonation** to remove odor-causing compounds (EPA, 2022).

1.5.2. Chemical Parameters

1.5.2.1 pH (Potential of Hydrogen)

pH is one of the most fundamental chemical parameters used to indicate the acidity or alkalinity of water, typically measured on a scale from 0 to 14 (APHA, 2017). pH is commonly determined using color indicators such as phenolphthalein and methyl orange, though modern water quality assessments rely more on digital pH meters and probes for accuracy (EPA, 2022). Because pH affects numerous chemical and biological processes, it is considered the first and most essential measurement in water quality evaluation (WHO, 2017).

Elevated pH levels can make water taste bitter and can promote the scaling and encrustation of pipes, boilers, and industrial equipment, leading to corrosion and increased toxicity in the water system (BIS 10500:2012). High pH also reduces the effectiveness of chlorine-based disinfectants, which is a concern for swimming pools and treated water supplies (WHO, 2017).

Several methods exist for measuring pH, including pH paper strips, liquid color indicators, and electronic pH meters. However, pH sensors with electrodes are considered the most reliable and accurate for field and laboratory assessments (APHA, 2017).

The pH value is interpreted as follows:

$$pH < 7.0 \rightarrow \text{Acidic}$$

$$pH = 7.0 \rightarrow \text{Neutral}$$

$$pH > 7.0 \rightarrow \text{Alkaline/basic}$$

According to drinking water standards:

Acceptable pH range: 6.5–8.5 (BIS 10500:2012; WHO, 2017)

Maintaining this range ensures water remains non-corrosive, aesthetically acceptable, and safe for human consumption.

1.5.2.2 Alkalinity:

Alkalinity is defined as the concentration of ions in water that can neutralize hydronium ions (H_3O^+), representing the water's ability to resist changes in acidity (APHA, 2017). It reflects the presence of acid-neutralizing constituents in water, primarily bicarbonates, carbonates, and hydroxides, which may originate from dissolved minerals, organic salts, dissolved gases, or other natural and anthropogenic sources (EPA, 2022).

Alkalinity does not directly indicate pH but instead shows how effectively water can buffer acids and maintain chemical stability—an essential factor for aquatic life and water treatment processes (WHO, 2017).

In simple terms, alkalinity acts as a buffer system, helping water maintain a stable pH even when acids are introduced.

Types of Alkalinities

1. **Bicarbonate Alkalinity (HCO_3^-)**

   The most common form in natural waters, dominant when pH is below 8.3 (APHA, 2017).

2. **Carbonate Alkalinity (CO_3^{2-})**

   Significant in waters where pH ranges between 8.3 and 10.

3. **Hydroxide Alkalinity (OH^-)**

   Present when pH exceeds 10, often in waters affected by industrial or chemical pollution (EPA, 2022).

Measurement of Alkalinity

Alkalinity is commonly measured by **acid titration** and reported in **mg/L as $CaCO_3$**.

- **Phenolphthalein Alkalinity (P-Alkalinity):**

  Measures hydroxide ions and half of carbonate alkalinity (endpoint pH 8.3).

- **Total Alkalinity (T-Alkalinity):**

  Measures the total acid-neutralizing capacity (endpoint pH 4.5).

These titration methods are widely included in standard water testing protocols such as APHA Standard Methods and BIS procedures.

Desirable Levels and Standards

According to BIS IS 10500:2012 for drinking water:

- **Desirable limit:** 200 mg/L as $CaCO_3$
- **Permissible limit:** 600 mg/L as $CaCO_3$ (in absence of an alternate source)

High alkalinity generally does not pose direct health risks but may influence water taste and increase sodium concentration—especially in softened water (WHO, 2017).

Effects of Alkalinity

1. **pH Stability**

   Alkalinity helps maintain consistent pH, protecting aquatic organisms from sudden acidity changes (UNESCO, 2020).

2. **Corrosion and Scaling**

   - High alkalinity → scaling in pipes, boilers, and industrial equipment.
   - Low alkalinity → corrosive water that can damage plumbing and metal surfaces

(EPA, 2022).

3. **Water Treatment**

   Alkalinity determines the required chemical dosage for coagulation, pH adjustment, and disinfection. It also affects chlorine efficiency and sedimentation processes (APHA, 2017).

4. **Aquatic Life**

   Adequate alkalinity acts as a natural buffer, preventing harmful pH fluctuations and supporting a stable aquatic ecosystem (WHO, 2017).

Major Components of Alkalinity

a. Carbonate Alkalinity

b. Bicarbonate Alkalinity

c. Caustic (Hydroxide) Alkalinity

1.5.2.3 Hardness:

The concentration of multivalent cations—primarily calcium (Ca^{2+}) and magnesium (Mg^{2+})—present in water is referred to as water hardness (APHA, 2017). Hardness occurs when water dissolves minerals from rocks and soil, particularly limestone, gypsum, and dolomite (WHO, 2017). Because groundwater remains in prolonged contact with mineral-bearing rocks, it typically has higher hardness than surface water (EPA, 2022).

Hard water affects daily household activities. For example, during bathing, hard water reduces soap lather formation because calcium and magnesium react with soap to form insoluble scum (UNESCO, 2020). Over time, untreated hard water can cause scaling inside pipes, water heaters, and industrial boilers, reducing efficiency and increasing maintenance costs (EPA, 2022). Hard water may also contribute to dry skin and hair, although this is not considered a direct health hazard (WHO, 2017).

According to BIS drinking water standards (BIS IS 10500:2012):

- Acceptable limit: 200 mg/L (as $CaCO_3$)
- Rejection limit: 600 mg/L (as $CaCO_3$), if no alternative source exists

Water hardness is commonly classified into:

a. Carbonate Hardness (Temporary Hardness)

Caused by the presence of bicarbonates and carbonates of calcium and magnesium. It can be removed by boiling or by lime-soda treatment (APHA, 2017).

b. Non-Carbonate Hardness (Permanent Hardness)

Caused by chlorides, sulfates, and nitrates of calcium and magnesium. This type cannot be removed by boiling but can be reduced through ion exchange or reverse osmosis (EPA, 2022).

Removal of Hardness:

Removing hardness results in soft water, which prevents scaling and improves soap lather formation. Common hardness removal methods include:

• Boiling (for temporary hardness)
• Lime-soda process
• Ion-exchange softening
• Reverse osmosis (RO)
• Electrodialysis

1.5.2.4 COD (Chemical Oxygen Demand):

It is the amount of oxygen required for the decomposition of both biodegradable and non-biodegradable organic matter present in the system (APHA, 2017; EPA, 2022). It is determined by adding $K_2Cr_2O_7$ and H_2SO_4 in the sample of water to be tested and noting the amount of $K_2Cr_2O_7$ left in the sample by titrating the water sample with ferrous ammonium sulphate using ferroin as indicator (APHA, 2017). $K_2Cr_2O_7$ is such an oxidizing reagent wherein, apart from the decomposition of organic matter, it also carries out decomposition of certain inorganic substances, resulting in higher oxygen demand than actual, which is also termed as Dichromate Demand (WHO, 2017).

1.5.2.5 BOD (Biochemical Oxygen Demand):

The amount of oxygen required for the decomposition of biodegradable organic matter present in the system is termed as BOD (WHO, 2017; EPA, 2022). BOD during 5 days at

20°C is taken as standard, and BOD is approximately 68% of the ultimate BOD (APHA, 2017). BOD is determined by diluting the known volume of the test sample with the aerated sample (APHA, 2017). Microorganisms like bacteria use organic matter as a source of food. When this material is metabolized, oxygen is consumed (UNEP, 2016). If this process takes place in water, the dissolved oxygen in a sample of the water will be consumed. If there is a substantial amount of organic matter in the water, high amounts of dissolved oxygen will be consumed in order to make sure that the organic matter decomposes (WHO, 2017). However, this creates problems since aquatic plants and animals require DO to survive. You can measure biological oxygen demand with the dilution method. If the BOD levels are high, the water is contaminated (CPCB, 2020).

1.5.2.6 Acidity:

Acidity of water refers to its capacity to neutralize a base and is primarily caused by the presence of dissolved carbon dioxide, mineral acids, and organic acids (APHA, 2017; EPA, 2022). It is an important parameter in assessing water quality, as high acidity can corrode pipes, harm aquatic life, and indicate pollution (WHO, 2017). The acidity level is commonly measured using pH, where lower pH values (below 7) signify higher acidity (BIS 10500:2012). Monitoring and controlling water acidity is essential for maintaining safe drinking water and a healthy aquatic environment (UNEP, 2016).

Types of Acidity

Acidity in water can be classified based on the substances responsible for it and the pH range at which it is neutralized. The main types are:

1. **Mineral Acidity (Strong Acidity):** Caused by strong acids such as hydrochloric acid (HCl), sulfuric acid (H_2SO_4), and nitric acid (HNO_3) (APHA, 2017; EPA, 2022). It is neutralized by a strong base like sodium hydroxide ($NaOH$) up to a pH of 4.5 (WHO, 2017).
2. **Carbon Dioxide Acidity (Weak Acidity):** Produced by dissolved carbon dioxide (CO_2) forming carbonic acid (H_2CO_3) in water (APHA, 2017; EPA, 2022). This type of acidity is weak and is neutralized by a base up to a pH of 8.3 (WHO, 2017).

3. **Organic Acidity:** Results from the presence of organic acids such as humic and fulvic acids, commonly found in natural waters due to decomposition of plant material (UNEP, 2016; APHA, 2017).
4. **Total Acidity:** The combined effect of both strong and weak acids present in the water (APHA, 2017; EPA, 2022). It is determined by titrating the sample to a specific endpoint pH (WHO, 2017).

Measurement of Acidity

The acidity of water is measured to determine its ability to neutralize bases (APHA, 2017; EPA, 2022). It helps assess water quality and detect the presence of acidic pollutants (WHO, 2017). The measurement is usually performed through acid–base titration and expressed in mg/L as $CaCO_3$ (calcium carbonate equivalent) (APHA, 2017).

Desirable Levels and Standards

There is no fixed permissible limit of acidity in drinking water by BIS or WHO, but it is generally expressed as mg/L of $CaCO_3$ (BIS 10500:2012; WHO, 2017).
• For most natural waters, acidity values are less than 5 mg/L as $CaCO_3$ (APHA, 2017).
• Values above 10 mg/L may indicate pollution from industrial effluents or acid rain (UNEP, 2016).

Effects of Acidity

Acidity in water can have significant impacts on human health, aquatic life, and infrastructure (WHO, 2017; EPA, 2022). The effects depend on the degree of acidity (pH level) and the duration of exposure (UNEP, 2016).

1. Effects on Human Health

- **Corrosive Action:** Acidic water can corrode pipes and plumbing systems, causing metals like lead, copper, and iron to leach into drinking water.
- **Health Hazards:** Consumption of such water may lead to metal poisoning, gastrointestinal irritation, and tooth enamel erosion.
- **Taste and Odor:** Acidic water often has a sour or metallic taste, making it unpleasant for drinking and cooking.

2. Effects on Aquatic Life

- **pH Sensitivity:** Most aquatic organisms thrive within a pH range of **6.5 to 8.5**.
- **Toxicity:** When pH falls below 5.5, fish and other aquatic species experience stress, reproductive failure, or death.
- **Metal Solubility:** Increased acidity raises the solubility of toxic metals (like aluminium and mercury), which can contaminate water and harm aquatic ecosystems.

3. Effects on Environment and Soil

- Acidic rainwater or runoff lowers soil pH, reducing nutrient availability and affecting plant growth.
- It may also damage crops and degrade forest ecosystems.

4. Effects on Infrastructure and Industry

• Acidic water corrodes metal pipes, tanks, and concrete structures, increasing maintenance costs (EPA, 2022; WHO, 2017).

• In industries like power plants and manufacturing, it can interfere with processes that require neutral or alkaline water (APHA, 2017; UNEP, 2016).

1.6 Effect Of Water Quality Parameter on Aquatic Life:

The effects of various water quality parameters like turbidity, alkalinity, hardness, and pH on aquatic life can vary depending on the specific organisms involved and the magnitude of the changes

1.6.1 Turbidity

Low Turbidity: Clear water allows sunlight to penetrate deeper, promoting photosynthesis in aquatic plants (UNEP, 2016; EPA, 2022). It also enables predators to spot their prey easily (WHO, 2017).

High Turbidity: Excessive turbidity can reduce light penetration, inhibiting photosynthesis and reducing oxygen production (APHA, 2017; UNEP, 2016). This can harm plants and other organisms that rely on them for food and habitat. Suspended particles can also clog the gills of fish and other aquatic organisms, affecting their respiration and feeding (EPA, 2022; WHO, 2017).

1.6.2 Alkalinity:

Low Alkalinity: Low alkalinity can result in acidic conditions, which may be harmful to some aquatic organisms, particularly those sensitive to changes in pH (WHO, 2017; UNEP, 2016). Acidic conditions can also leach heavy metals from sediments, posing additional risks (EPA, 2022).

High Alkalinity: High alkalinity can buffer against pH changes, which can be beneficial for organisms sensitive to pH fluctuations (APHA, 2017). However, extremely high alkalinity levels can still have negative effects on some species, altering their physiological processes (WHO, 2017).

1.6.3 Hardness:

Low Hardness: Low hardness may lead to softer water, which can affect the ability of organisms to maintain proper ion balance and may interfere with shell formation in some species, such as mollusks and crustaceans (WHO, 2017; UNEP, 2016).

High Hardness: High hardness can provide essential minerals for aquatic organisms and may contribute to shell formation in some species. However, excessively hard water can also lead to scaling in pipes and equipment and may impact aquatic life that prefers softer conditions (EPA, 2022; APHA, 2017).

1.6.4 pH Value:

Low pH: Acidic conditions (low pH) can be harmful to many aquatic organisms, affecting their physiology and behavior (WHO, 2017; UNEP, 2016). Acidic water can also release metals into the water, which can be toxic to aquatic life (EPA, 2022).

High pH: Alkaline conditions (high pH) can also be detrimental, especially to organisms that require lower pH levels. High pH can affect the availability of nutrients and may lead to toxicity issues with certain chemicals (APHA, 2017; WHO, 2017).

In summary, maintaining balanced water quality parameters is essential for the health of aquatic ecosystems. Abrupt or significant changes in turbidity, alkalinity, hardness, or pH can stress or harm aquatic organisms, disrupting the delicate balance of these ecosystems (UNEP, 2016; EPA, 2022). Monitoring and managing these parameters are critical for the conservation and protection of aquatic life (WHO, 2017).

- **Sahni, P. & Varshney, D. (2024)-**Investigated that river water has been used as a source of drinking water. River water is now a source of hydropower generation, irrigation, aquaculture, navigation, transportation, and supports socio-economic activity, human settlement and aquatic living forms. Due to increase in industrialization, urbanization, mining, sewage disposal and rampant use of technology, Yamuna River water is becoming polluted day by day. It is threatening the survival of life itself. The present study focuses on physico-chemical analysis of Yamuna River water at Poiya Ghat site, Agra to determine its quality and pollution profile over a period of 8 months. The results show the high concentration of some parameters like Total Hardness, TDS, Fluoride content, BOD and COD than the WHO and BIS permissible limits for drinking water. Other parameters like pH, EC, Chloride content, DO were under the permissible limits of WHO and BIS. . Heavy metals like Zn, Cr, Ni, Cd, Mn and Cu were very low in concentration while Pb and Fe were high in concentration as compared to the WHO and BIS permissible limits. Physico-chemical analysis shows the high level of pollution in Yamuna River at Poiya Ghat site of Agra. This may possess a health risk to several rural communities as well as urban localities who rely on the river water primarily as their source of domestic water and irrigation. The present study showed an urgent need for a continuous pollution monitoring and treatment programmes of Yamuna River water in India.

- **Yadav, D. & Pandey, G. (2021)-** Gorakhpur is situated from bank of Rapti River. Rapti River is joined at Sohgaura with Ami River which is highly polluted as compare to Rapti River. Rapti River has a confluence into Ghaghra River at Kaparwar Ghat. The present study is conducted to determine the water quality of Rapti River using physical and Chemical parameters including pH, Turbidity, Total dissolved solids, Dissolved Oxygen (DO), and Total Hardness by collecting Three sets of samples from five locations from December, 2020 to February, 2021. The results obtained by standard methods were compared with water quality criteria prescribed by Indian Standard/Central Pollution Control Board. The study also reveals that the water quality of Rapti River is highly deteriorated due to anthropogenic activities such as urbanization, construction activities, agricultural activities, discharge of untreated sewage and disposal of solid wastes into river.

- **Mishra, KD (2020)-** This paper illustrates the quality of Yamuna River water from the Agra region of Uttar Pradesh. This river is the second largest tributary of Ganga which flows from Uttrakhand to Haryana and then enters into Noida, Uttar Pradesh, and lastly merges into Ganga River at Triveni Sangam in Prayagraj, Uttar Pradesh. This river is highly polluted due to industrial waste, agricultural waste, and fertilizers. During the investigation, the researcher tried to find out the Physico-chemical characteristics of Yamuna River water for five months. Six Sampling sites were selected in the Agra region.

- **Gour, S. & Gour, M. (2019)-** Rivers are most important recourse for consumption as freshwater for human being. The quality of water resource can be measured and identified by its physicochemical and biological parameters. This paper presents a study of water quality of River Narmada through analysis of some of its selected physicochemical and biological parameters. This study show that quality of river water is not favorable and goes down day by day due to domestic and industrial effluents mixing in the river at town. The finding of this study gives definitely an alert message to water resource managers for the improvement of water quality management by proper treatment and monitoring the River.

- **Kumar, S. & S. (2018)-** This study was completed around kushmhi railway station area during December 2016 to March 2017, and evaluate the some Groundwater quality parameters of chose India Mark-II and Shallow depth hand pumps. Total 52 water samples collected from different stations and analysis of the various physical, chemical parameters such as Temperature, pH, Turbidity, Total hardness, Total alkalinity, Turbidity, Fluoride, Chloride. The results also compared with 10500-2012 standards. The analysis reveals that most of the parameter is found under acceptable limit, but some are exceeding the acceptable limit of BIS standards. The present study is based on the results of the ground water is alkaline nature and some stations water samples contaminated for selected parameters, and exhibits that the some shallow depth hand pumps are affected by high extent of physico-chemical parameters except chloride in study area.

3.1 METHODOLOGY:

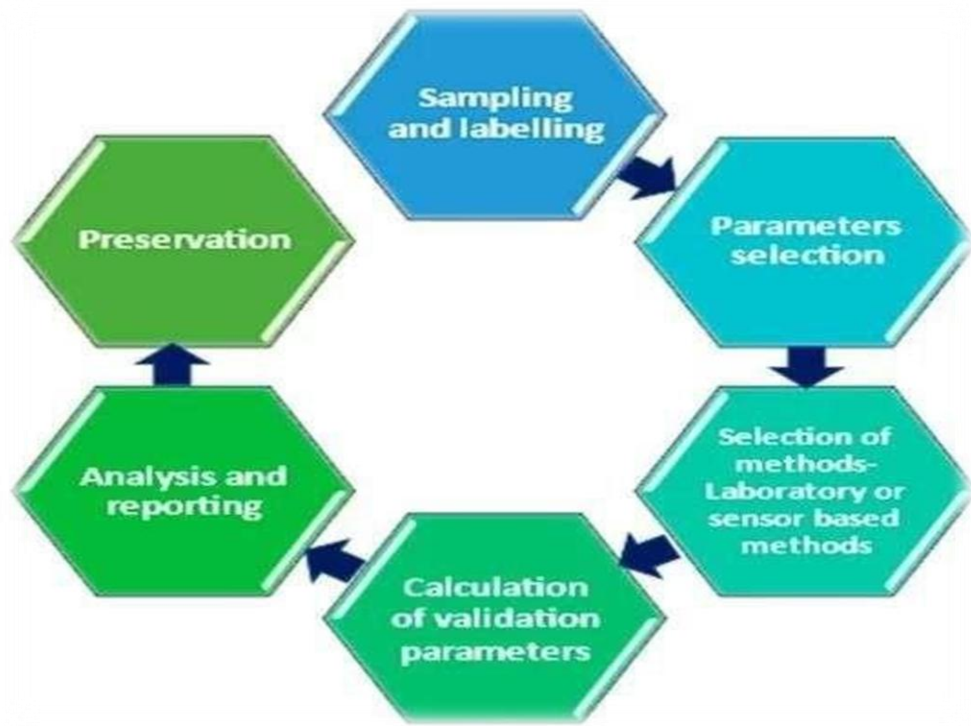


Fig 3.1: Flow Chart of Methodology

3.2 MAP VIEW AND SAMPLES COLLECTION:

We have selected four places which is present is Gorakhpur, GIDA for testing their properties.

Fig.3.2: Rajghat

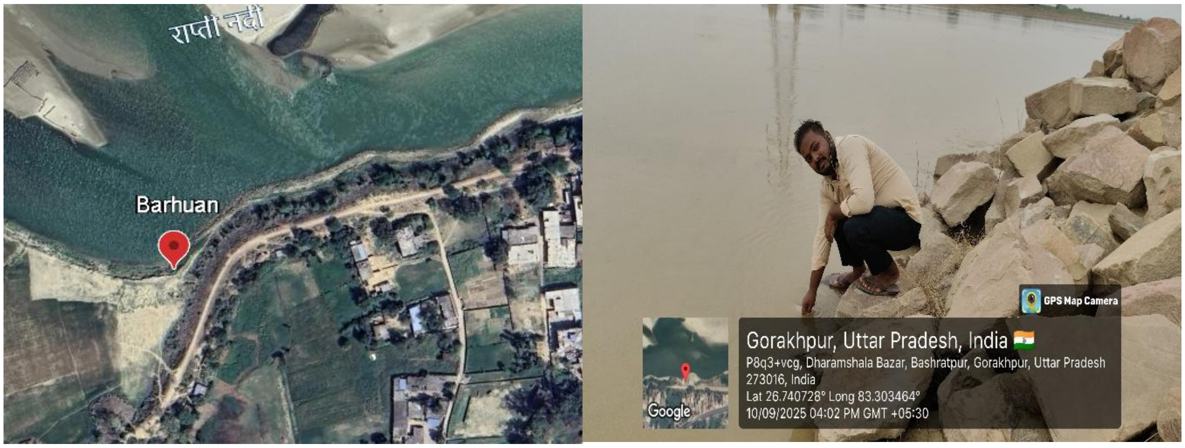


Fig.3.3 : Barhuan

Fig.3.4: Mokshdham

Fig.3.5: Muhamadpur Maphi

3.3 Experimental Procedure for Different Parameters

3.3.1 pH

3.3.1.1 About pH

The pH of water samples was detected with the help of pH strips present in the environmental lab. PH strips are pieces of paper that change colour depending on the pH – the acidity or alkalinity – of a liquid. PH strips are a cheap and relatively accurate way of measuring the pH of ant liquid. pH indicators are usually weak acids or weak bases that change colour at specific pH's. Color changes occur when the acid or base accepts or donates a proton. When this happen, it depends upon the chemical specific characteristics. Some indicators are more complex and they have more than one color change. The paper is treated with a chemical known as Flavin which is soluble in water and changes color in the presence of various types of solutions. It is coated with a chemical indicator that transitions in the presence of hydrogen or hydroxide ions. pH stands for the "power of hydrogen". The pH of water is defined as the log of the reciprocal of hydrogen ion concentration present in that water. The logarithmic scale of pH indicates that as pH increases, the H+ concentration will decrease by a power of 10 i.e., each number below 7 is 10 times more acidic than the previous number when counting down. Likewise, when counting up above 7, each number is 10 times more basic than the previous number. Thus, at a pH of 0, H+ has a concentration of M. At a pH of 7, this decreases to 0.0000001 M. At a pH of 14, there is only 0.000000000000001 M H+. If the pH of water is 7, it is neutral. If pH of water is less than 7, it is acidic and if the pH of water is more than 7, it is basic.

3.3.1.2 PROCEDURE OF pH

1. Gather materials (Litmus paper, Sample of water, clean & dry Beaker or test tube)
2. Take Sample in Beaker and dip the litmus paper into the solution
3. Take a strip of blue litmus paper and a strip of red litmus paper.
4. Dip the blue litmus paper into the sample If it turns red the substance is acidic
5. Dip the red litmus paper into the sample If it turns blue, the substance is alkaline (basic)
6. Compare the colour change to a pH scale or chart to estimate the pH level of substance.

3.3.1.3 ENVIORNMENTAL SIGNIFICANCE OF pH

If water whose pH levels are less than 7 is consumed in excess quantity, it may increase the acidity of the mouth which can cause the demineralization of tooth enamel which in turn can lead to tooth decay. Any liquid with a pH of 10 or more can cause burns depending on the tissue

it touches and how long the tissue is exposed. pH values greater than 11 can cause skin and eye irritations, as does a pH below 4. A value below 2.5 will cause irreversible damage to skin and organ linings. pH can also affect the solubility and toxicity of chemicals and heavy metals in the water. Lower pH levels increase the risk of mobilized toxic metals that can be absorbed, even by human. High pH levels can have negative impact on gastrointestinal system. In addition to that, pH levels outside of 6.5-9.5 can damage and corrode pipes and other systems, further increasing heavy metal toxicity. If the pH of water is too high or too low, the aquatic organisms living within it will die.

Acceptable range of pH – It lies between 6.5 – 8.5

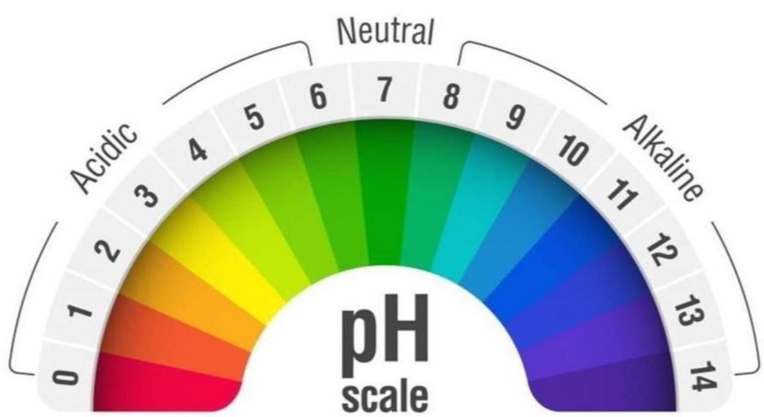


Fig.3.6: pH Scale

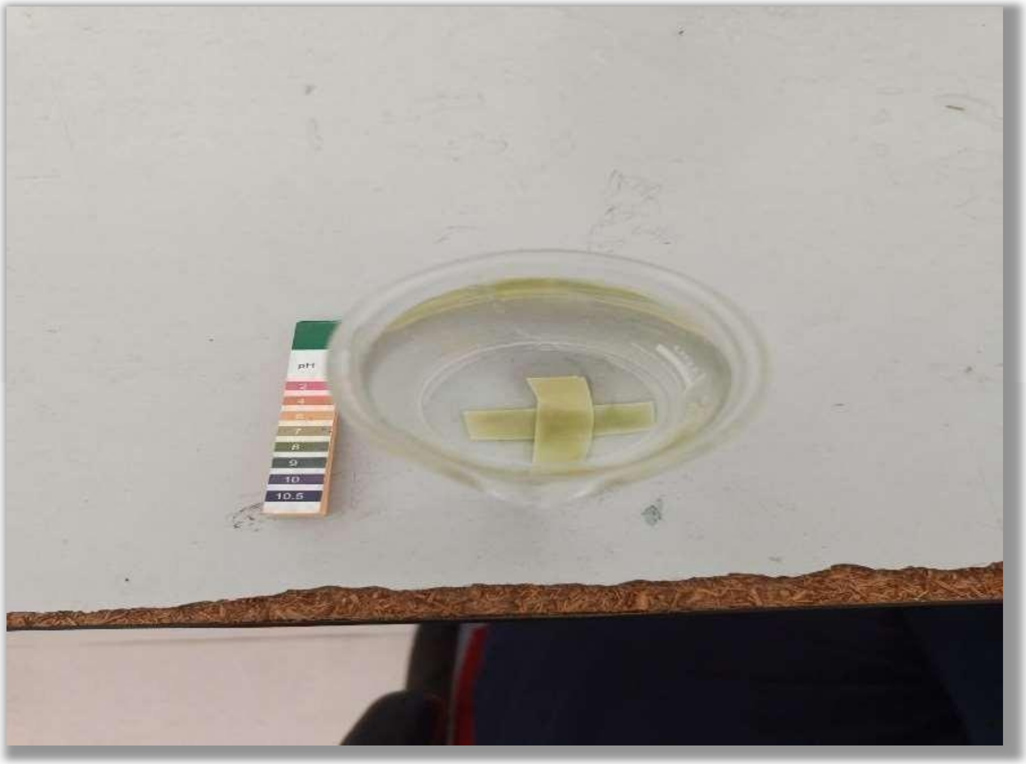


Fig.3.7: Analysis of pH by litmus paper

Fig.3.8: Analysis of pH-by-pH meter

3.3.2 ALKALINITY

3.3.2.1 About Alkalinity

It is the measure of ability of water to neutralize acids. The major portion of alkalinity in natural water is caused by carbonates, bicarbonates and hydroxides. It affects the boilers by forming scales on it. Alkalinity is the water's capacity to resist changes in pH that would make the water more acidic. It also helps protect your health and piping when it comes to drinking water. The term "buffering capacity" usually denotes this capacity. Alkalinity also refers to the capability of the water to neutralize the acid. This is true that it has a buffering capacity. A buffer is a solution in which acid can be added without changing the concentration of available H+ ions (as in changes 30 in pH). It absorbs the excessive H+ ions and protects the water from fluctuations in pH. The pH and the alkalinity level of the well water can be affected by various factors such as acidic sanitizers, rain, fill water, and other product applications that can change the alkalinity level over time. The majority of alkalinity in the surface water is from calcium carbonate, CaCo3, leached from soil and rocks. This process is enhanced if the soil and rocks have been broken up for various reasons like urban development and mining. Limestone contains high levels of calcium carbonate. When used to decrease the acidity at homes, it can runoff into surface waters and raise the alkalinity level.

3.3.2.2 Apparatus and Chemical Required

Apparatus – Burette stand, Burette, conical flask, measuring cylinder. Reagents – Titrant Standard Sulfuric acid (N/50) Indicator – phenolphthalein, Water sample and Sodium Thiosulphate.

3.3.2.3Experimental procedure

1. Fill the burette to HSo4 Solution
2. Take a 25ml water in flask. Add few drop of phenolphthalein indicator
3. Note the Initial reading on burette scale Titrate against H_2 sou till the pink color disappear
4. Note the end point reading and get volume of used' H2So4 in ml.
5. Add 1-3 drop of methyl orange in same sample flask
6. Titrate it, till the appearance of light orange color
7. Note down the final reading and the volume of used H2So4

a) Repeat the steps of using the sample to get correct value.

b) Calculate the total alkalinity of sample

3.3.2.4 Environmental significance of alkalinity

Drinking too much alkaline water may disrupt the body's normal pH. This can lead to a condition called metabolic alkalosis, which may cause confusion, nausea, vomiting, hand tremors, muscle twitching, and tingling in the face, hands or feet. Fish and other aquatic life require a pH range of 6.0 to 9.0, and because alkalinity buffers against rapid pH changes, it protects the living organisms who require a specific pH range. Higher alkalinity levels in surface water will buffer acid rain and other acid wastes, preventing pH changes that are harmful to aquatic life. Alkalinity is also in important considering the treatment of wastewater and drinking water because it influences cleaning processes such as anaerobic digestion. Water may also be unsuitable for use in irrigation if the alkalinity level in the water is higher than the natural level of alkalinity in the soil.

Fig 3.9: Analysis of Alkalinity

Fig.3.10: Alkalinity Test

3.3.3Hardness of Water

3.3.3.1About Hardness

Hardness in water is that characteristic which prevents the formation of sufficient lather with soap. The hardness is usually caused by the presence of calcium and magnesium salts present in the water which form scum by reaction with soap Thus, hard water contains dissolved magnesium and calcium ions which make it more difficult for the water to form a lather with soaps. Dissolved magnesium ions and calcium ions can get into the water when it comes into contact with limestone and other rocks that contain calcium compounds. There are 2 types of hardness of water: Temporary hardness or carbonate hardness: It is caused by the bicarbonates and carbonates of calcium and magnesium. It can be removed by boiling. Permanent hardness or non-carbonate hardness: It may be caused by the sulphates and chlorides of calcium and magnesium. It can be removed by ion exchange process, etc. It is expressed in milligrams of calcium carbonate.

Water hardness refers to the concentration of divalent metal cations, primarily calcium (Ca^{2+}) and magnesium (Mg^{2+}), in water. These ions are responsible for the formation of scale and poor soap lathering. Hardness is not a health hazard but is a critical indicator of water quality for domestic, agricultural, and industrial use.

It is usually expressed in milligrams per liter (mg/L) as calcium carbonate ($CaCO_3$).

Types of Hardness

1. **Temporary Hardness (Carbonate Hardness)**:

- Caused by bicarbonates of calcium and magnesium.
- Can be removed by boiling, which precipitates the bicarbonates.

2. **Permanent Hardness (Non-carbonate Hardness)**:

- Caused by chlorides, nitrates, and sulfates of calcium and magnesium.
- Cannot be removed by boiling; requires chemical treatment (e.g., ion exchange or lime softening).

3.3.3.2 Experimental Procedure

1. The burette is filled with Standard EDTA Solution to 0 level
2. Take 100ml sample water in flask. If sample having high calcium content then take smaller Volume and dilute to 50ml.

   Add 1 ml Ammonia buffer

1. Add 1 to 2 drop of Eriochrome Black T Indicator

2. The Solution twins into wine red colour

- Note down Initial Reading
- Titrate the content against EDTA solution. At the end point colour change from wine red to blue Colour.

• Note the final Reading and record it. Repeat the process till we get concordant value

Formula Used;

$$N_1 * V_1 = N_2 * V_2$$

Where,

N_1=Normality of known solution

V_1=Volume of known solution

N_2= Normality of unknown solution

V_2= Volume of unknown solution Acceptable limit: 200 mg/L Permissible limit: 600 mg/L

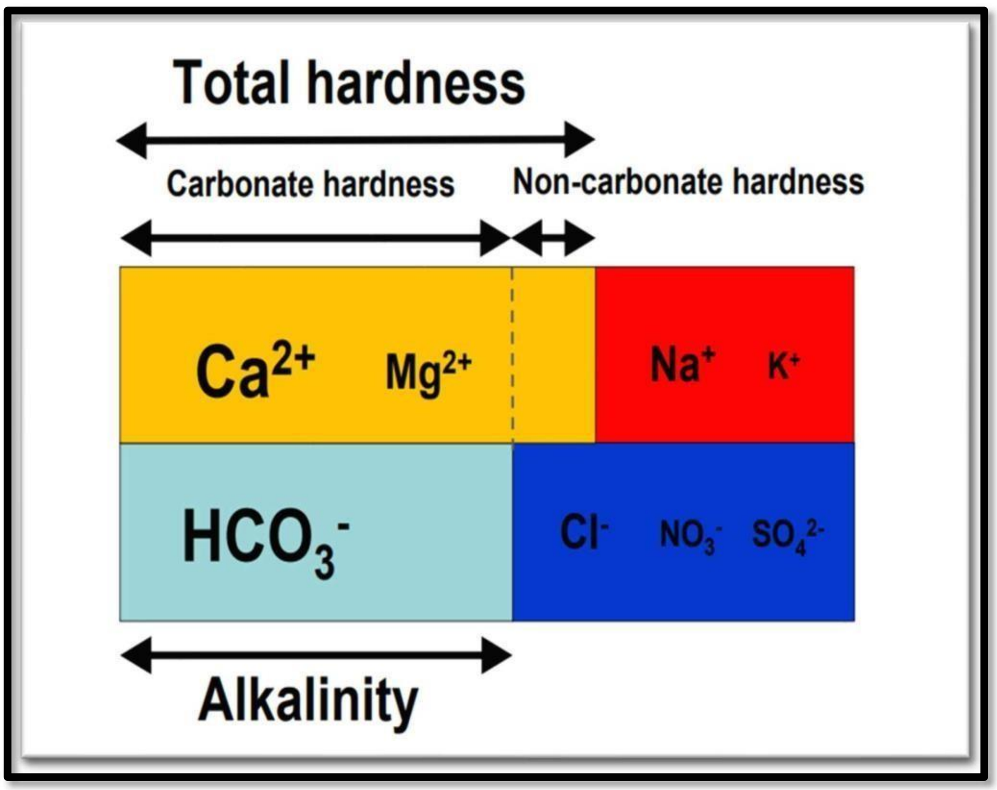


Fig.3.11: Alkalinity & Hardness of water

Fig 3.12: Analysis of hardness

Fig.3.13: Hardness test

3.3.4Turbidity
3.3.4.1 About Turbidity

Turbidity is the cloudiness or haziness of a fluid caused by large numbers of individual particles that are generally invisible to the naked eye, similar to smoke in air. The measurement of turbidity is a key test of water quality. Turbidity may be caused by particles suspended or dissolved in water that scatter light making the water appear cloudy or murky. Particulate matter can include sediment - especially clay and silt, fine organic and inorganic matter, soluble coloured organic compounds, algae, and other microscopic organisms. The suspended particles absorb heat from the sunlight, making turbid waters become warmer, and so reducing the concentration of oxygen in the water. The suspended particles scatter the light, thus decreasing the photosynthetic activity of plants and algae, which contributes to lowering the oxygen concentration even more. As a consequence of the particles settling to the bottom, shallow lakes fill in faster, fish eggs and insect larvae are covered and suffocated, gill structures get clogged or damaged.

Turbidity is a key physical parameter that indicates the clarity of water and the amount of suspended particles present. It is defined as the measure of light-scattering ability of water due to the presence of suspended solids such as clay, silt, organic matter, plankton, and microscopic

organisms. The more particles present in water, the higher its turbidity, and the cloudier it appears. Turbidity is usually measured in Nephelometric Turbidity Units (NTU) using a device called a nephelometer or turbidimeter, which assesses how much light is scattered at a 90-degree angle by particles in the water sample. Clear water has low NTU values (typically below 1 NTU), while highly turbid water may exceed 100 NTU or more, depending on pollution levels.

3.3.4.2 Experimental Procedure

1. Switch on the Turbidity meter (at least 30 min. before the Test

Prepare 400 NTU Solution

2. Calibrate the turbidity meter to 400 NTU using the Standard solution by adjusting

the calibration knob.

3. Calibrate the turbidity meter to 0 NTU using Distilled water and by adjusting the calibration knob.
4. Read the Turbidity meter by inserting the sample**.**

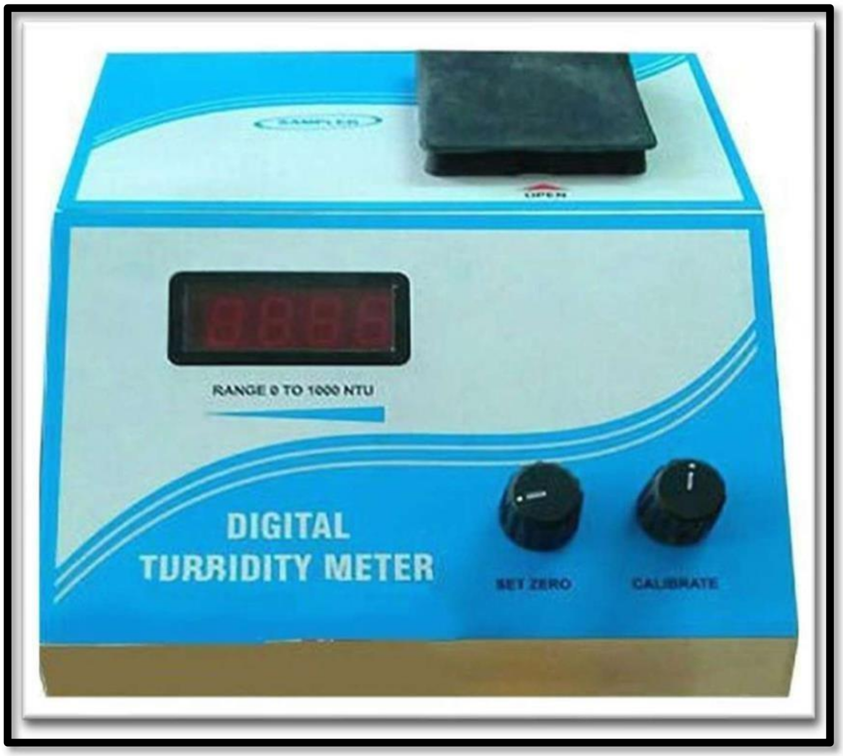


Fig.3.14: Turbidity Meter

3.3.5 Acidity

3.3.5.1 About Acidity

In any sample of pure water, a small number of water molecules are constantly reacting with each other. In this reversible reaction, one water molecule donates a proton (H^+) to a neighboring water molecule. This process forms a hydronium ion (H_3O^+) and a hydroxide ion ($OH^–$).

The chemical equation for this equilibrium is:

$$\mathbf{2H_2O(l) \rightleftharpoons H_3O^+(aq) + OH^-(aq)}$$

The presence of the hydronium ion is what gives water its acidic properties, while the hydroxide ion gives it basic properties.

Ion Product Constant and pH

The concentrations of hydronium and hydroxide ions in water are linked by the ion product constant for water (K_w). At 25°C, K_w has a value of 1.0×10^{-14}.

$$K\_w = [H_3O^+]\,[OH^-] = 1.0 \times 10^{-14}$$

In absolutely pure water, the concentrations of H_3O^+ and OH^- are equal, creating a neutral solution. We can find their concentrations:

$$[H_3O^+] = [OH^-] = \sqrt{(1.0 \times 10^{-14})} = 1.0 \times 10^{-7}\ M$$

The pH scale provides a more convenient way to measure acidity and is defined as the negative logarithm of the hydronium ion concentration.

$$pH = -\log[H_3O^+]$$

For pure water at 25°C, the pH is:

$$pH = -\log(1.0 \times 10^{-7}) = 7$$

A pH of 7 is the hallmark of a neutral solution.

How Acidity Changes

The neutral balance in water is disturbed when other substances are dissolved in it.

- Acids are substances that increase the concentration of H_3O^+ ions in water. This leads to a decrease in the concentration of OH^- ions and results in a pH value below 7. For example, when carbon dioxide (CO_2) from the atmosphere dissolves in rainwater, it forms carbonic acid (H_2CO_3), making rain naturally slightly acidic.
- Bases are substances that increase the concentration of OH^- ions. This causes the H_3O^+ concentration to decrease and results in a pH value above 7.
- In essence, while we call water neutral, its constant, slight ionization is the very foundation of the pH scale and a crucial factor in the chemistry of acids and bases.

3.3.5.2 Types of Acidity

In water analysis, two types of acidity are commonly measured:

1. **Mineral Acidity (Methyl Orange Acidity):**

This is caused by strong mineral acids like sulfuric acid and hydrochloric acid. It is determined by titrating the sample to a pH of approximately 3.7, which is the endpoint for the methyl orange indicator.

2. **Total Acidity (Phenolphthalein Acidity):**

This includes both strong mineral acids and weak acids, such as carbonic acid (from dissolved CO_2)and acetic acid. It is determined by titrating the sample to a pH of 8.3, the endpoint for the phenolphthalein indicator. For most natural water samples, measuring total acidity is sufficient.

3.3.5.3 Apparatus and reagents Required

- Burette (50 mL) with a stand
- Pipette (25 mL or 50 mL)
- Conical flasks (Erlenmeyer flasks) (250 mL)
- Beakers
- Graduated cylinders
- Funnel
- Wash bottle with distilled water
- Reagents:
- Water Sample: The water to be tested.
- Standard Sodium Hydroxide (NaOH) Solution (0.02 N)

3.3.5.4 Experimental Procedure

- Prepare the Burette:
- Rinse a clean 50 mL burette with a small amount of the 0.02 N NaOH solution.
- Fill the burette with the 0.02 N NaOH solution, ensuring there are no air bubbles in the tip.
- Record the initial volume reading on the burette.
- Prepare the Water Sample:
- Using a pipette, accurately measure 50 mL (or a suitable volume) of the water sample into a clean 250 mL conical flask.
- Add the Indicator:
- Add 2-3 drops of phenolphthalein indicator solution to the water sample in the conical flask. If the sample is acidic, the solution will remain colorless.
- Perform the Titration:
- Place the conical flask under the burette on a white tile or surface to clearly see the color change.

- Slowly add the NaOH solution from the burette to the water sample while constantly swirling the flask to ensure mixing.
- Continue adding the NaOH drop by drop as you get closer to the endpoint. The endpoint is reached when the colorless solution turns a faint but persistent pink color that lasts for at least 30 seconds.
- Stop the titration at the first sign of this persistent pink color.
- Record the Result:
- Record the final volume reading from the burette.
- The volume of NaOH used is the final reading minus the initial reading.

Fig.3.15: Acidity Analysis

3.3.6 Total Dissolved Solid

3.3.6.1 About TDS

Total Dissolved Solids (TDS) is a measure of the total concentration of all inorganic and organic substances dissolved in a given volume of water. These substances can include minerals, salts, metals, cations, and anions. Essentially, TDS provides a quantitative measure of the amount of dissolved components in water, besides the pure H_2O molecules themselves. The TDS value is typically expressed in milligrams per liter (mg/L) or parts per million (ppm).

Composition of TDS

The composition of TDS in water can be diverse and varies depending on the source of the water. However, the principal constituents are generally inorganic salts. These include:

- Cations:
- Calcium (Ca^{2+})
- Magnesium (Mg^{2+})
- Sodium (Na^{+})
- Potassium (K^{+})
- Anions:
- Carbonate (CO_3^{2-})
- Bicarbonate (HCO_3^{-})
- Chloride (Cl^{-})
- Sulfate (SO_4^{2-})
- Nitrate (NO_3^{-})

In addition to these common ions, TDS can also include small amounts of organic matter, as well as trace metals such as iron, manganese, and copper.

3.3.6.2 Measurement of TDS

There are two primary methods for measuring TDS in water:

1. **Gravimetric Method**: This is the most accurate method and is typically performed in a laboratory. It involves filtering a known volume of water, evaporating the water, and then weighing the remaining residue.

   Apparatus:

- Filtering apparatus with filter paper
- Evaporating dish
- Drying oven

2. **Analytical balance**

   Electrical Conductivity (EC) Method: This is a more common and convenient method, especially for on-site testing. It utilizes the fact that the electrical conductivity of water is directly related to the concentration of dissolved ionized solids. A TDS meter measures the EC and then converts it to an estimated TDS value.

Apparatus:

TDS Meter: This is a handheld, pen-like device with electrodes that are dipped into the water sample. The meter provides a direct reading of the TDS level.

In summary, Total Dissolved Solids are a crucial parameter in assessing water quality. While a certain level of dissolved minerals is normal and can even be beneficial, high concentrations of TDS can indicate water quality problems that may have environmental and health implications. Regular monitoring of TDS levels is essential for ensuring the safety and suitability of water for various uses

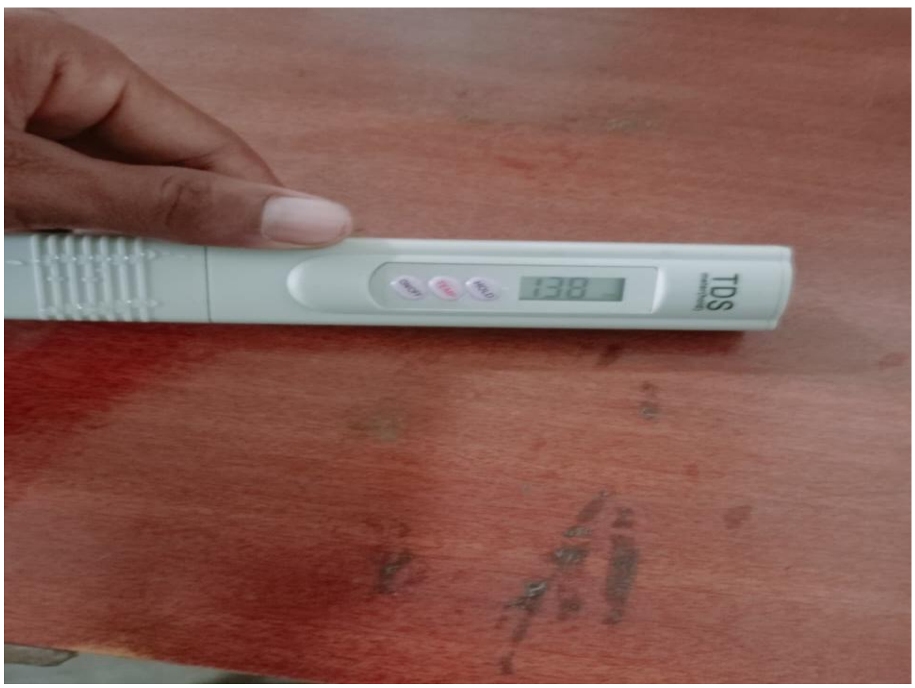


Fig.3.16: TDS Analysis

3.3.7 Normalized Difference Turbidity Index

3.3.7.1 About NDTI

In the context of water, NDTI stands for Normalized Difference Turbidity Index. It is a remote sensing index used to estimate the turbidity of water bodies. Turbidity refers to the cloudiness or haziness of a fluid caused by large numbers of individual particles that are generally invisible to the naked eye, similar to smoke in the air. The measurement of turbidity is a key test of water quality.

The NDTI is calculated using the spectral reflectance values of water pixels obtained from satellite imagery. The principle behind it is that clear water has a higher electromagnetic reflectance in the green spectrum compared to the red spectrum. Conversely, turbid water, which contains suspended particles like sediment and algae, has a higher reflectance in the red part of the spectrum.

The formula for calculating NDTI is:

Where:

- Red is the reflectance value in the red band of the electromagnetic spectrum.
- Green is the reflectance value in the green band of the electromagnetic spectrum.

The resulting NDTI values typically range from -1 to +1.

- Higher positive values (closer to +1) generally indicate higher turbidity, meaning the water is more cloudy or contains more suspended particles.
- Lower or negative values (closer to -1) suggest lower turbidity and clearer water.

Applications and Significance of NDTI in Water Management

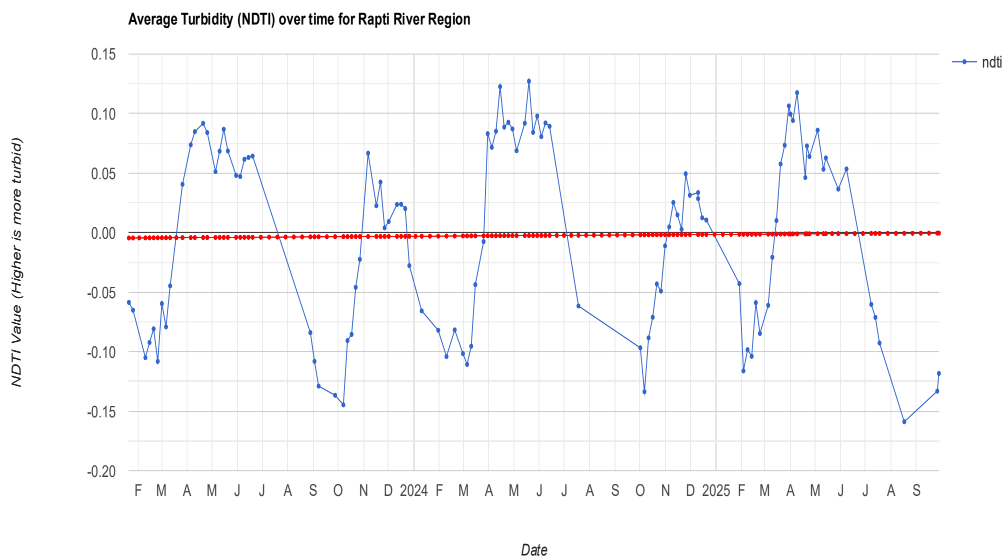
The NDTI is a valuable tool for monitoring and managing water resources, offering several advantages over traditional water sampling methods, which can be time-consuming and provide limited spatial coverage.

Key applications include:

- **Water Quality Monitoring:** NDTI provides a synoptic view of turbidity levels over large areas, such as lakes, reservoirs, coastal waters, and rivers. This allows for the identification of pollution hotspots, algal blooms, and areas with high sediment loads.
- **Environmental Monitoring:** Changes in turbidity can be an indicator of environmental stressors, such as soil erosion from land-use changes, industrial discharge, or runoff from agricultural areas. NDTI can be used to track these changes over time.
- **Wetland Management:** The health of wetland ecosystems is crucial for biodiversity and water quality. NDTI helps in monitoring the turbidity in these sensitive environments, which can be affected by both natural and human-induced factors.
- **Sedimentation Monitoring:** In reservoirs and dams, sedimentation is a significant issue that can reduce water storage capacity. NDTI can be used to monitor the rate and extent

of siltation.

- **Coastal and Estuarine Studies:** NDTI is useful for studying the dynamics of suspended sediments in coastal zones and estuaries, which is important for understanding coastal erosion and the health of marine ecosystems.
- In summary, the Normalized Difference Turbidity Index is a powerful remote sensing tool that provides a cost-effective and efficient way to assess and monitor the turbidity of water bodies over large areas. This information is crucial for effective water resource management, environmental protection, and scientific research**.**



Graph 3.1: NDTI Analysis by google earth engine

CHAPTTER 04
RESULT AND DISCUSSION

We select 4 sites which is located at Gorakhpur, GIDA and at each site we select Points and then we select levels of water for collection of samples for Testing. We have visited 4 sites in 6th semester sites visited noted the following parameters which are as follows;

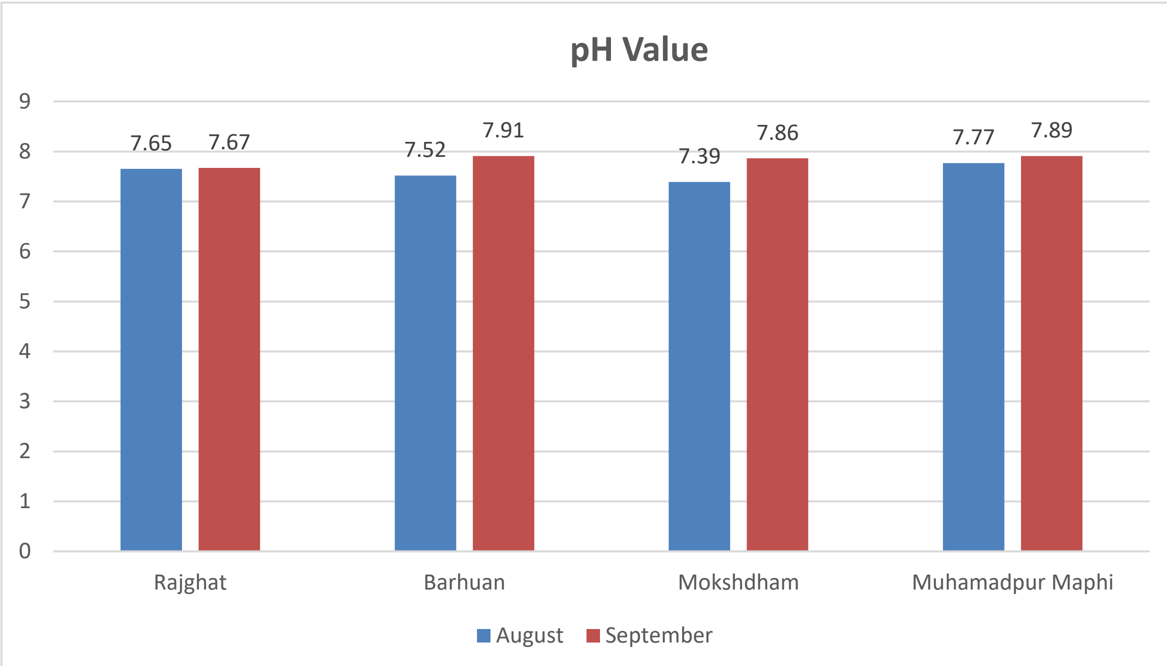
- Turbidity
- Hardness of Water
- pH value
- Alkalinity
- Acidity
- TDS

Standard Value of Water Quality Parameter Specified by BIS or WHO for drinking water:

Table 4.1: Water Quality Parameters specified by BIS and WHO

PARAMETER	UNIT	BIS 1991	WHO 2011	ANALYTICAL METHOD
Turbidity	NTU	1	1	Digital Turbidity Meter
Hardness of water	ppm	300	600	Titration
pH value	---	6.5-7.5	6.5-7.5	Universal Indicator
Alkalinity	ppm	200	600	Titration
Acidity	ppm			Titration
TDS	ppm	50-300	300-600	Titration

4.1 pH Test Result:-

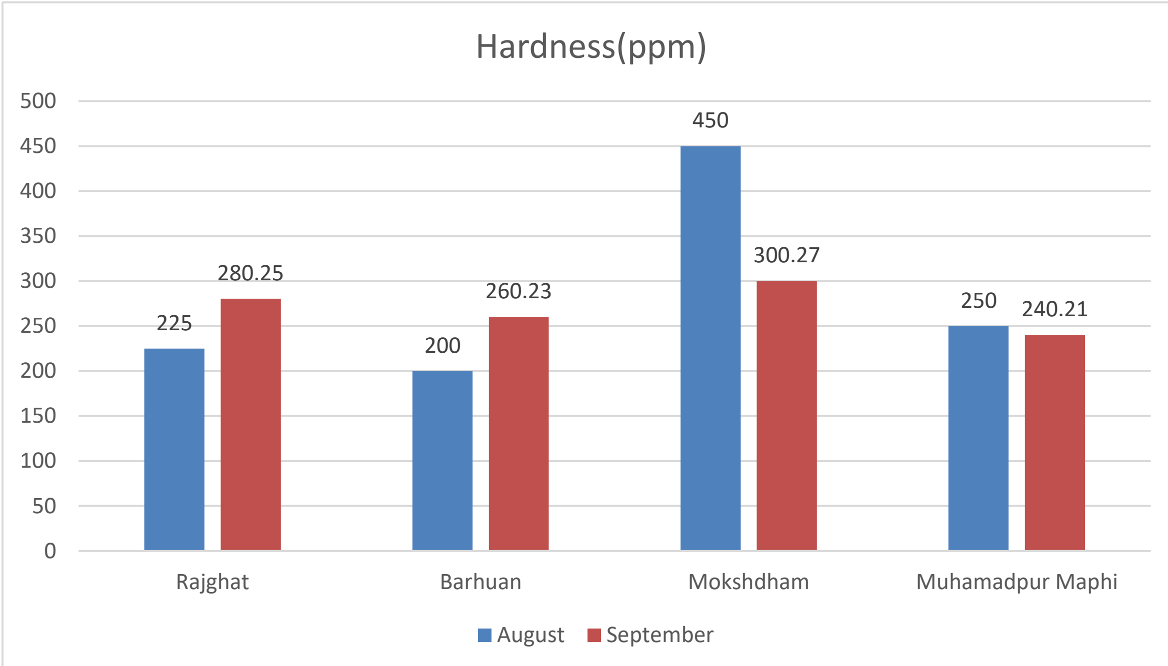


Graph4.1: pH value across different location

Table 4.2: pH Test Result

Month	Rajghat	Barhuan	Mokshdham	Muhamadpur Maphi
August	7.65	7.52	7.39	7.77
September	7.67	7.91	7.86	7.89

4.2 Hardness Test Result:-

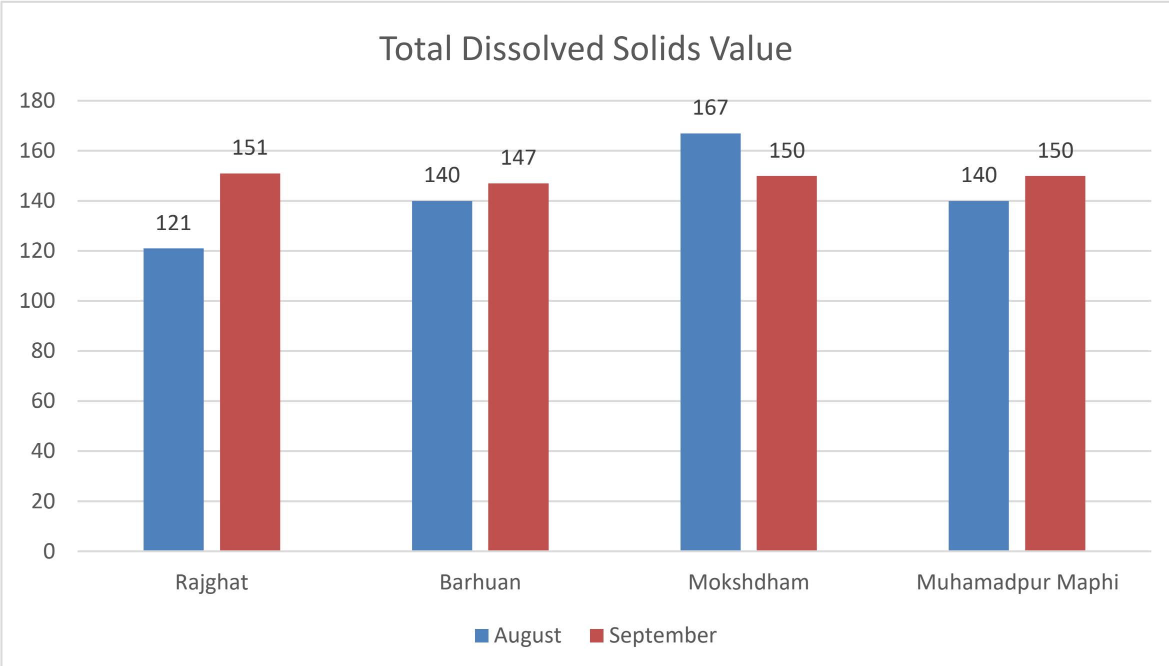


Graph 4.2:- Hardness across different location

Table 4.3: Hardness test Result

Month	Rajghat	Barhuan	Mokshdham	Muhamadpur Maphi
August	225	200	450	250
September	280.25	260.23	300.27	240.21

4.3 TDS test Result

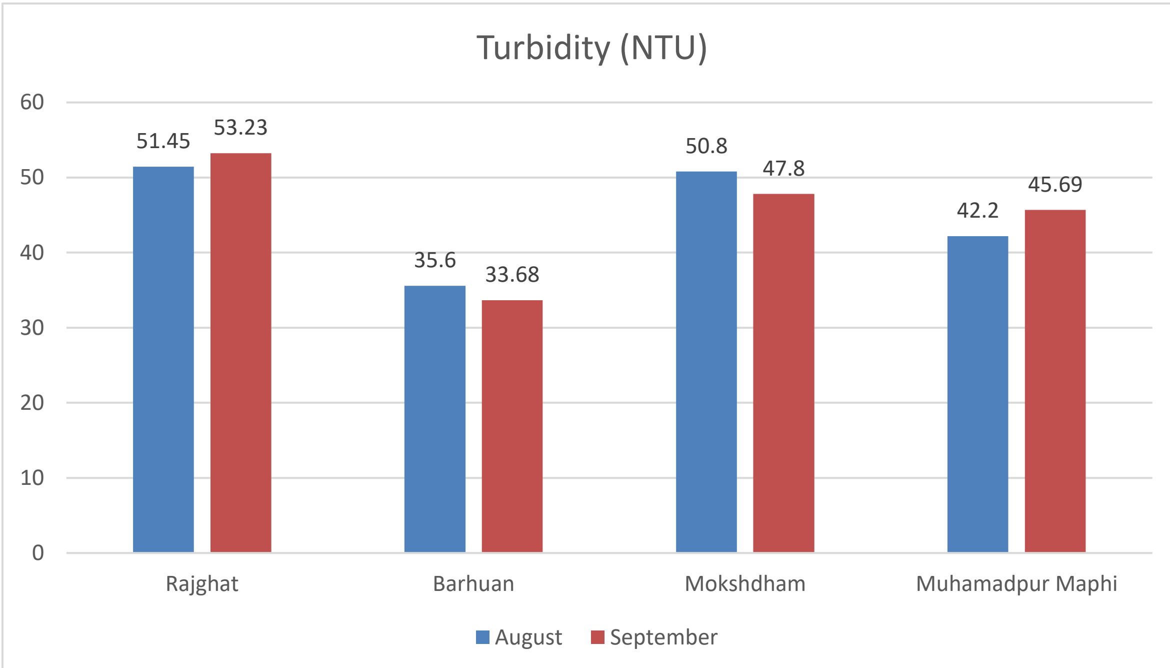


Graph 4.3: TDS across different location

Table 4.4: Total Dissolved solids Test Result

Month	Rajghat	Barhuan	Mokshdham	Muhamadpur Maphi
August	121	140	167	140
September	151	147	150	150

4.4: Turbidity test result:-

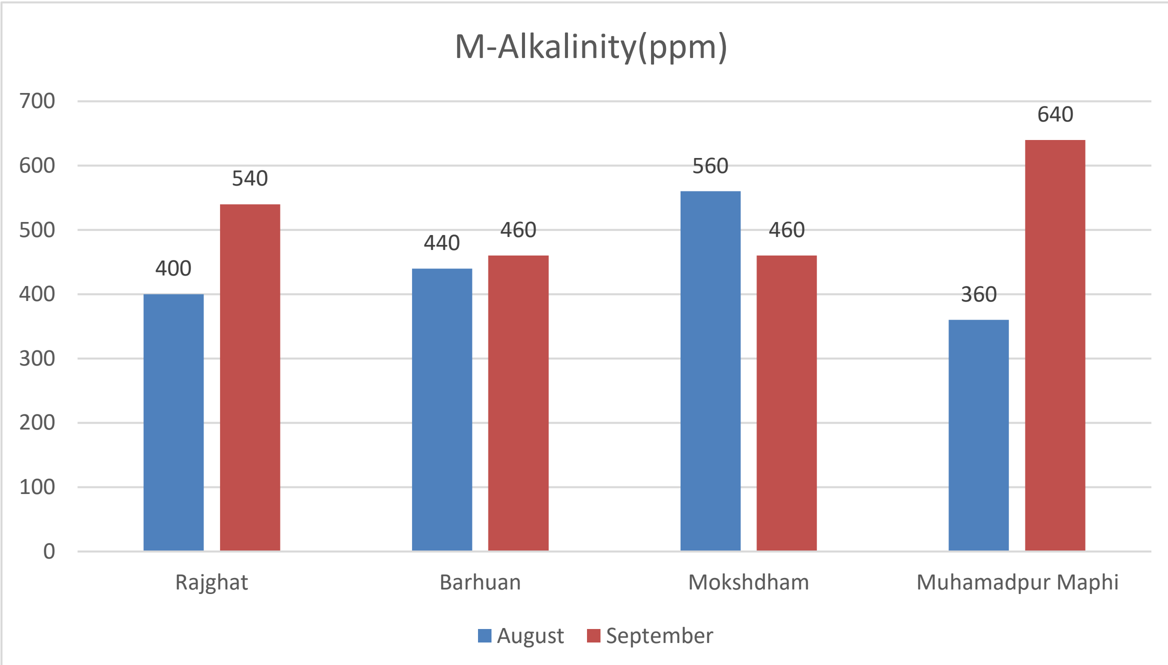


Graph 4.4: **Turbidity across different location**

Table 4.5: Turbidity test result

Month	Rajghat	Barhuan	Mokshdham	Muhamadpur Maphi
August	51.45	35.6	50.8	42.2
September	53.23	33.68	47.8	45.69

4.5: Alkalinity Test Result:-

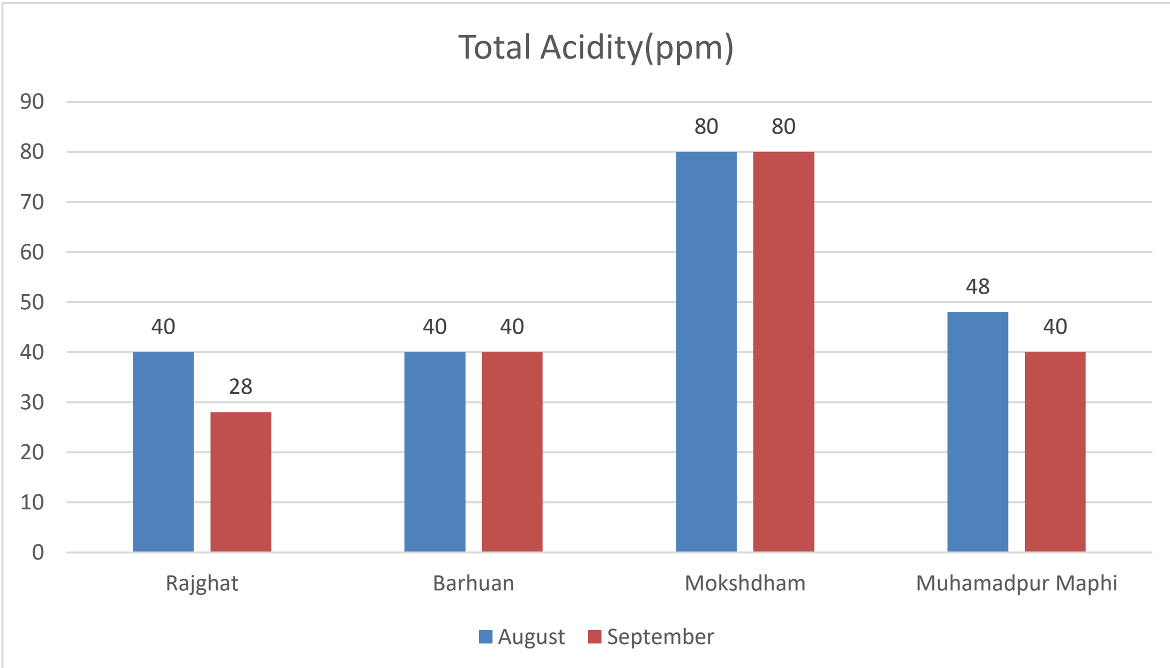


Graph 4.5: Alkalinity across different location

Table 4.6: Alkalinity Test Result

Month	Rajghat	Barhuan	Mokshdham	Muhamadpur Maphi
August	400	440	560	360
September	540	460	460	640

4.6: Acidity test Result: -



Graph 4.6: -Acidity across different location

Table 4.7: -Acidity Test result

Month	Rajghat	Barhuan	Mokshdham	Muhamadpur Maphi
August	40	40	80	48
September	28	40	80	40

CHAPTER 05
CONCLUSION

The assessment of water quality parameters in the Rapti River revealed significant variations in physical, chemical, and biological characteristics. The results indicate that the river water is polluted with high levels of turbidity, total suspended solids, biochemical oxygen demand, chemical oxygen demand, and total coliform.

The pH, hardness, alkalinity, and chloride content were found to be within the acceptable limits for drinking water, as per the Bureau of Indian Standards. However, the turbidity and bacterial contamination exceeded the permissible limits, posing a risk to human health.

The study highlights the need for effective wastewater management, conservation of natural habitats, and implementation of sustainable agricultural practices to mitigate the pollution of the Rapti River. Regular monitoring and assessment of water quality parameters are essential to ensure the river's water is safe for drinking, irrigation, and other uses.

The study concludes that the Rapti River is facing severe pollution threats due to anthropogenic activities. The high levels of pollutants in the river water pose significant environmental and

health risks. Immediate measures are required to control pollution and restore the water quality of the Rapti River. These measures may include:

- Implementation of efficient wastewater treatment systems

The assessment of water quality parameters of the Rapti River provides essential insights into the current status of river health and the potential impacts of natural and human-induced activities on this important freshwater resource. Based on the analysis of various physical (such as temperature, turbidity, color, TDS, taste, and odor) and chemical parameters (such as pH, alkalinity, hardness, chloride, etc.), it can be concluded that while some water quality indicators fall within the acceptable limits prescribed by BIS and WHO, others may exceed permissible thresholds in specific stretches of the river, especially in areas close to urban settlements, agricultural zones, or industrial activity.

Elevated values of parameters like turbidity, chloride, or total hardness may be attributed to sewage discharge, agricultural runoff, or industrial effluents, highlighting the need for effective wastewater management and pollution control strategies. Parameters such as alkalinity and pH indicate the buffering capacity of the water and the overall chemical stability of the river.

Continuous and seasonal monitoring of the river's water quality is crucial to detect temporal trends, identify pollution sources, and implement corrective actions before irreversible ecological damage occurs. Public awareness, stricter enforcement of environmental regulations, and community involvement in river conservation can collectively improve the health of the Rapti River.

In conclusion, this study emphasizes the importance of regular water quality monitoring, the need for sustainable water resource management, and the urgent requirement for pollution mitigation to ensure that the Rapti River remains a viable and safe water source for future generation.

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LIST OF PUBLICATION

1. Srivastav, S., Jaiswal, K., Yadav, A., Singh, D. P., & Shukla, R. "Comprehensive Analysis of Water Quality of Rapti River in Gorakhpur Region". Presented at the 1st National Conference on Sustainable Civil Engineering & Geospatial Technologies: Materials & Innovations for Climate Resilience (NCRT-SCE 2026), held on 15–16 April 2026.

2. Srivastav, S., Jaiswal, K., Yadav, A., Singh, D. P., & Shukla, R. "Comprehensive Analysis of Water Quality of Rapti River in Gorakhpur Region". Published in International Research Journal of Engineering and Technology (IRJET), Volume 13, Issue 03, March 2026, E-ISSN: 2395-0056, P-ISSN: 2395-0072.

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CL-1, Sector-7, GIDA, Gorakhpur, Uttar Pradesh, India

Comprehensive Analysis of Water Quality of Rapti River in Gorakhpur Region

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Abstract: The project titled "Comprehensive Analysis of Water Quality of Rapti River in Gorakhpur Region" The Rapti River, a vital water resource for the ecosystem and populace of the Gorakhpur region, faces severe environmental stress from escalating anthropogenic pressures. This proposal outlines a comprehensive study to analyze the spatio-temporal variations in the river's water quality. The primary objective is to create a detailed assessment of its hydro-chemical status, identify major pollution hotspots, and quantify the impact of key pollution sources .This research will involve systematic water sample collection from strategic locations, including upstream (baseline) sites, confluence points with major tributaries (like the Ami River), and areas downstream of significant discharge points (such as urban drains and the Gorakhpur Industrial Development Authority - GIDA). The analysis will focus on a suite of critical physico-chemical and biological parameters, including but not limited to: pH, Turbidity, Total Dissolved Solids (TDS), Dissolved Oxygen (DO), Biochemical Oxygen Demand (BOD), Chemical Oxygen Demand (COD), Total Hardness, nitrates, phosphates, and fecal coliform (MPN).The study hypothesizes that the river quality deteriorates significantly as it passes through the Gorakhpur urban agglomeration, with untreated domestic sewage and industrial effluents being the principal contributors. Particular attention will be given to the confluence with the heavily polluted Ami River, which is suspected to be a critical hotspot for chemical and heavy metal contamination.

***Keywords**: Physico-chemical parameters, pollution hotspots ,Biochemical Oxygen Demand(BOD), Chemical Oxygen Demand (COD),Dissolved Oxygen(DO).*

Certificate No : BIT/NCRT-SCE/2026/ 25

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Branch: Civil Engg. has presenting the paper/abstract titled Comprehensive Analysis of Water quality of Rapti River in Gorakhpur Region.

the **"National Conference on Sustainable Civil Engineering & Geospatial Technologies: Materials & Innovations for Climate Resilience (NCRT-SCE 2026)"** held from 15th to 16th April, 2026.

We congratulate the participant for outstanding performance and contribution.

Mr. Pratish Kannaujiya	**Mr. Ankur Kumar**	**Mr. Baijnath Nishad**	**Prof. (Dr.) Roop Ranjan**
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Comprehensive Analysis of Water Quality of Rapti River in Gorakhpur Region

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Abstract - *The Rapti river is an important freshwater resource for drinking, irrigation and ecological equilibrium. Yet, this river is facing growing pollution problems due to anthropogenic activities, industrial effluents and agricultural runoffs. This research focuses on the assessment of physio-chemical parameters of the river water to understand the water quality. Parameters like pH, turbidity, hardness, alkalinity, chloride etc. were measured at some sites of the river. We suggest regular monitoring, efficient wastewater treatment, conservation of natural habitats and sustainable agricultural practices to avert pollution and to ensure the river water be potable and suitable for irrigation and other uses. This article underlines the necessity of integrated water resource management towards sustainable development and prevention of water quality degradation of Rapti River.*

Key Words: Rapti River, water quality of Gorakhpur, pollution, environmental assessment, water parameters, water management, river pollution.

1. INTRODUCTION

The Gorakhpur district depends on the Rapti River because it delivers fundamental water supplies which support agricultural activities, industrial operations, and household needs. Human activities have created significant pollution threats which endanger the river. The Rapti River pollution affects both environmental systems and public health because it harms aquatic ecosystems and the human populations that depend on the river. The river starts in the lower Himalayas and goes into the Gangetic plains, which supports diverse biological systems while serving as the main water supply for nearby communities.

1.1 OBJECTIVE OF STUDY

Comprehensive water quality assessment requires investigation of rising human activities which include urban development and agricultural expansion and industrial growth Water quality evaluation determines whether water can be used for drinking, irrigation, and industrial purposes. This study aims to analyze selected water quality parameters at key locations along the Rapti River to assess the spatial variation in water quality and pinpoint sources of pollution.

1.2 KEY WATER QUALITY PARAMETERS

Water quality parameters which include pH, dissolved oxygen (DO), turbidity, temperature, biochemical oxygen demand (BOD), chemical oxygen demand (COD), nitrates phosphates and microbial content help determine aquatic system health and pollutant levels. The parameters enable identification of contamination sources while they monitor changes in time and assist water management policies.

2. LITERATURE REVIEW

- Sahni, P. & Varshney, D.1 (2024)- River water has served as an essential resource since ancient times because people have used it for drinking water and irrigation and hydropower generation and aquaculture and navigation and transportation. The area supports economic development while providing habitats for human communities and sustaining various aquatic organisms. The Yamuna River now faces severe pollution problems because industrialization and urbanization and mining activities and sewage disposal and technological advancements have increased pollution levels. The contamination of the river system creates serious health hazards for both urban and rural populations who use it as their primary water source for household needs and agricultural purposes. The research study demonstrates that continuous pollution monitoring and effective treatment programs need to be established as urgent requirements for restoring and protecting the water quality of the Yamuna River in India.

- Yadav, D. & Pandey, G.3 (2021)- The city of Gorakhpur sits on the banks of the Rapti River which flows into the Ami River at Sohgaura because the Ami River carries higher pollution levels than the Rapti River. The Rapti River continues its course until it reaches the Ghaghra River at Kaparwar Ghat. Researchers conducted a study to evaluate the water quality of Rapti River through measurements of physical and chemical parameters which included pH and turbidity and total dissolved solids and dissolved oxygen and total hardness. The research results show that human activities which include urban development and construction work and agricultural practices and discharge of untreated sewage and disposal of solid waste lead to significant contamination and environmental degradation of the river ecosystem.

- Pandit, P. & Fulekar, M.H.5 (2017)- The study assessed water quality across nine coastal regions of Gujarat, India, during the pre-monsoon season by measuring various physicochemical parameters. Principal Component Analysis (PCA) served as the statistical method that researchers used to evaluate and interpret their collected data. The results provide fundamental data about environmental conditions while showing how human activities affect coastal ecosystem health. This research shows that PCA functions effectively as a data simplification tool which enables researchers to extract essential information about water quality for improved environmental monitoring and management in Gujarat coastal areas.

- Rai, A.4 (2015)- The Ami River shows higher pollution levels than the Rapti River which harms groundwater quality in the surrounding regions. Three study sites were established at Kauriram which is located before the Ami meets the Rapti River and at Malav which is located before the Ami River and at Sohagaura which is located after their confluence to examine water quality from shallow and deep hand pumps. The researchers conducted an analysis of physical properties and chemical composition and biological characteristics of groundwater which they collected from Indian Mark hand pumps situated near both rivers. The results demonstrate that pollution from the Ami River deteriorates groundwater quality in Sohagaura, which shows that better river water management practices are necessary to safeguard local water resources.

- Alam, JB et al.2 (2007)- Researchers tested water quality from multiple locations along the Surma River which flows between Chattak and Sunamganj during both the dry season and the monsoon season. The river section contains two important industrial facilities which include a paper mill and a cement factory that could affect the water quality of the river. The analysis showed that people should not consume the river water in this area because it needs treatment to become safe for drinking. The water maintains sufficient quality for multiple surface water applications which demonstrates the requirement for effective management methods and pollution prevention measures to safeguard this important resource.

3. METHODOLOGY

3.1 Sample collection: We have collected water sample from four places which are present in GIDA, Gorakhpur, for testing their properties.

- Rajghat
- Barhuan
- Mokshdham
- Muhamdpur Maphi

Fig.3.1: Rapti River (Rajghat)

3.2 Selected Parameters: The water samples were analyzed for various physico-chemical parameters, including:

- pH
- Alkalinity
- Hardness
- Turbidity
- Acidity
- Total Dissolved Solids

3.3 Parameters Analyzed:

a) pH:

pH is a measure of the acidity or basicity of a solution. In the context of water quality, pH is an important parameter to measure.

- In water quality monitoring, pH is measured to:
- Determine the acidity or basicity of the water
- Assess the potential for corrosion or scaling
- The acceptable pH range for drinking water is typically between 6.5 and 8.5.

Fig.3.2: pH Test

b) Alkalinity:

Alkalinity is the ability of water to resist changes in pH when acids or bases are added. It's a measure of the concentration of bases, such as bicarbonate (HCO_3^-), carbonate (CO_3^{2-}), and hydroxide (OH^-), that can neutralize hydrogen ions (H^+). Alkalinity levels can vary depending on the water source and location.

Low alkalinity: Less than 10 mg/L as $CaCO_3$ (calcium carbonate)

Moderate alkalinity: 10-50 mg/L as $CaCO_3$

High alkalinity: Greater than 50 mg/L as $CaCO_3$

The acceptable limit of alkalinity in drinking water is 200 mg/L as $CaCO_3$ (calcium carbonate), according to IS 10500-2012 [1]. However, in the absence of an alternative source, the permissible limit can be extended to 600 mg/L as $CaCO_3$ [2].

c) Hardness:

Water hardness refers to the concentration of dissolved minerals, specifically calcium and magnesium, in water. These minerals can cause scaling, which can lead to problems in plumbing, appliances, and industrial equipment. Water hardness is typically measured in:

1. mg/L (milligrams per liter): This unit measures the concentration of calcium and magnesium ions in water.

2. ppm (parts per million): This unit is equivalent to mg/L.

3. DGH (degrees of general hardness): This unit measures the concentration of calcium and magnesium ions in water, with 1 DGH equivalent to 17.9 mg/L.

4. Grains per gallon (GPG): This unit measures the concentration of calcium and magnesium ions in water, with 1 GPG equivalent to 17.1 mg/L.

The acceptable limit for water hardness varies depending on the intended use:

Drinking water: 200-400 mg/L (11.3-23.4 GPG)

Industrial water: 100-500 mg/L (5.7-29.2 GPG)

High water hardness can cause:

Scaling: The formation of mineral deposits in pipes and appliances.

Corrosion: The degradation of pipes and equipment due to the presence of minerals.

Soap scum formation: The formation of soap scum on skin and surfaces.

Fig.3.3: Hardness Test

d) Turbidity: Turbidity refers to the cloudiness or haziness of water due to the presence of suspended particles, such as sediments, clay, silt, and other impurities. Turbidity can affect the appearance, taste, and odor of water, as well as its safety for drinking and other uses.

Turbidity is typically measured in:

Nephelometric Turbidity Units (NTU): This is the most common unit of measurement for turbidity.

Formazin Turbidity Units (FTU): This unit is similar to NTU but uses a different calibration standard.

WHO (World Health Organization): 5 NTU

USEPA (United States Environmental Protection Agency): 0.3-1 NTU

BIS (Bureau of Indian Standards): 5 NTU

Fig.3.4: Turbidity Test

4. STANDARD VALUE:

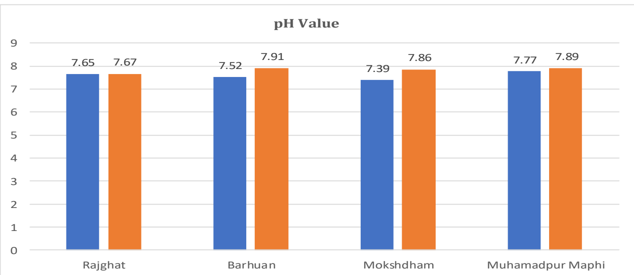
Table 4.1: Water Quality Parameters Specified by BIS and WHO

PARAMETER	UNIT	BIS 1991	WHO 2011	ANALYTICAL METHOD
Turbidity	NTU	1	1	Digital Turbidity Meter
Hardness	mg\l	300	600	Titration
pH value	---	6.5-7.5	6.5-7.5	Universal Indicator
Alkalinity	mg\l	200-600	200-600	Titration
Acidity	mg\l	50	50	Titration
TDS	mg\l	300-600	500	TDS Meter

5. RESULT & ANALYSIS

5.1 pH Test Result: -

	Rajghat	Barhuan	Mokshdham	Muhamadpur Maphi
Aug.	225	200	450	250
Sep.	280.25	260.23	300.27	240.21



Graph 5.1: pH values across different locations

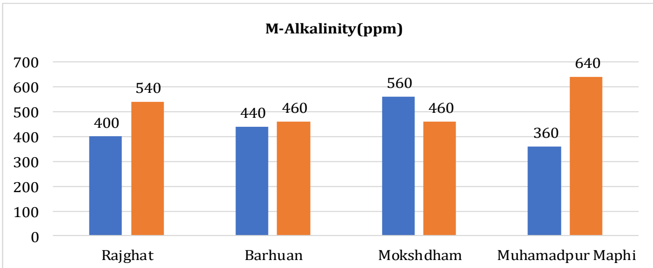
Table 5.1: pH Values

	Rajghat	Barhuan	Mokshdham	Muhamadpur Maphi
Aug.	7.65	7.52	7.39	7.77
Sep.	7.67	7.91	7.86	7.89

5.2 Alkalinity Test Result: -

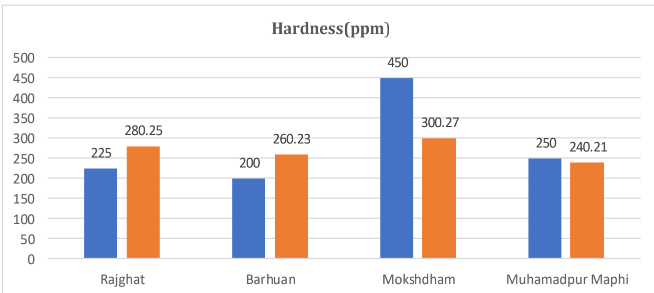
Table 5.2: Alkalinity Values

	Rajghat	Barhuan	Mokshdham	Muhamadpur Maphi
Aug.	400	440	560	360
Sep.	540	460	460	640



Graph 5.2: -Alkalinity across different locations

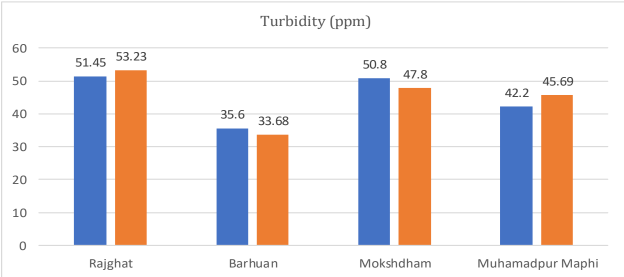
5.3 Hardness Test Result: -



Graph5.3: Hardness across different locations

Table 5.3 Hardness level

	Rajghat	Barhuan	Mokshdham	Muhamadpur Maphi
Aug.	51.45	35.6	50.8	42.2
Sep.	53.23	33.68	47.8	45.69

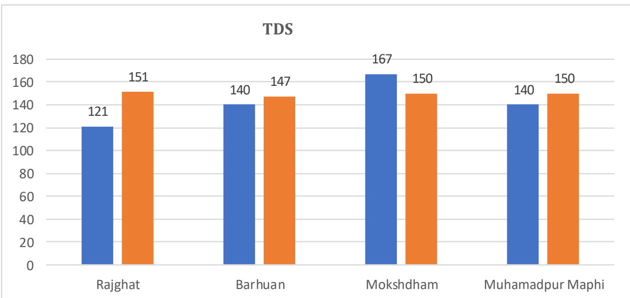


Graph 5.4: Acidity across different locations

Table 5.4: Acidity Values

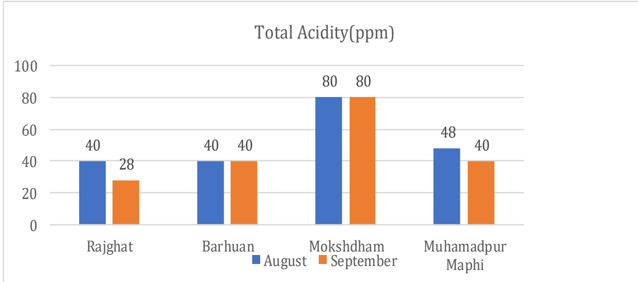
	Rajghat	Barhuan	Mokshdham	Muhamadpur Maphi
Aug.	40	40	80	48
Sep.	28	40	80	40

5.5 Turbidity Test Result: -



Graph 5.5: Turbidity across different locations

5.6 TDS Test Result: -



Graph 5.6: TDS across different location

Table 5.2: TDS Values

	Rajghat	Barhuan	Mokshdham	Muhamadpur Maphi
Aug.	121	140	167	140
Sep.	151	147	150	150

6. CONCLUSION

The evaluation of water quality parameters in the Rapti River indicated substantial fluctuations in physical, chemical, and biological attributes. The findings show that the river water is polluted because it has high levels of turbidity, total suspended solids, biochemical oxygen demand, chemical oxygen demand, and total coliform. The Bureau of Indian Standards says that the pH, hardness, alkalinity, and chloride levels were all within the acceptable range for drinking water. But the turbidity and bacteria levels were too high, which could be bad for people's health The Rapti River needs our help because we have to manage wastewater in a way, take care of natural habitats and use farming methods that do not harm the environment. We have to check the water of the Rapti River all the time to make sure it is safe for people to drink for farmers to use for irrigation and for things. The Rapti River is in trouble because of things that people have done. The water of the Rapti River has a lot of things in it which is very bad for the environment and for people's health. We need to do something now to stop the pollution of the Rapti River and make its water clean again. These measures may include: Implementation of efficient wastewater treatment systems

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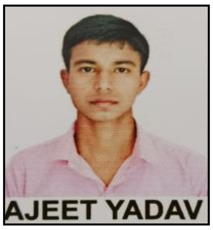

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